# A Theory of Optimal Disaster Insurance under Intermediary Constraints

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Koen van den Bosch*

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**Abstract**

Implicit insurance subsidies in housing markets protect households from rising weather damage but constrain insurers financially, undermining the long-run availability of insurance. Using a dynamic housing model calibrated to the California housing market, I characterize the optimal time-consistent insurance allocation that protects housing values while limiting insurance market crashes after disaster shocks. The severity of these crashes is governed by the economy-wide marginal propensity to insure. I decentralize the social planner's optimal allocation with an insurance subsidy whose magnitude varies with household leverage and the likelihood and cost of binding insurer capital constraints in the future. The model reveals that: (i) while a subsidy on insurance is optimal in most states, its magnitude lowers when high insurance demand coincides with a high probability of future insurer net worth collapses, (ii) a regulator influencing insurance prices directly reduces welfare as it generates persistent under-insurance. I further provide empirical evidence consistent with insurer financing constraints amplifying insurer withdrawals from insurance markets.

**JEL Codes:** G22, Q54, R31, G28, D62

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*Erasmus University, Rotterdam School of Management. Email: vandenbosch@rsm.nl.

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# 1 Introduction

Property insurance markets are under stress due to growing economic damages from natural disasters that create record-level claim payouts. Following disaster losses, the availability of insurance has declined as private insurers started to exit disaster-prone areas. Such insurance contractions make homeowners more exposed to natural disasters, affect house prices (Eastman et al., 2024) and potentially impose significant macroeconomic costs (ECB-EIOPA, 2023).

To keep insurance available and affordable, U.S. regulators have intervened across several dimensions, including by capping rate increases, providing insurance through residual insurers to homeowners who cannot obtain coverage in the private market, using state guaranty funds to pay claims when insurers default, and directly subsidizing insurance. While mitigating under-insurance, these policies could crowd out the long run availability of private insurance by undermining insurers' financial conditions.[1]

A fundamental question is therefore which regulation optimally balances the short-term need to keep homeowners insured with insurers' financial stability, a condition for keeping homeowners insured in the long run. The literature has made progress in estimating the sign and magnitude of regulatory distortions in insurance markets, yet lacks a structural analysis of (i) the conditions under which regulatory interventions are justified and (ii) how optimal regulation depends on the joint state of the insurance and housing markets.

In this paper, I develop a dynamic housing model with insurance intermediaries, calibrated to the California housing market, to characterize the optimal time-consistent insurance policy that jointly mitigates under-insurance and insurance market crashes. Two occasionally binding constraints are central to the model's inefficiencies that motivate regulatory intervention: a household collateral constraint and an insurer capital constraint. Disaster shocks destroy collateral and can lead to binding collateral constraints when insurance take-up is low. Upon binding collateral constraints, house prices drop sharply as households need to deleverage, destroying borrowing capacity and ultimately consumption. Similarly, disaster shocks lead to sharp reductions in the availability of insurance when the insurer's capital constraint binds, as insurers need to deleverage.

In a decentralized equilibrium (DE), households take house prices and insurer net worth as given. Consequently, they do not internalize how insurance decisions today affect next-

[1] Oh et al. (2021) and Liu and Liu (2024) show that insurance rate caps are associated with an increase in insurer exits. Abramson et al. (2025) show that underpriced government policies can crowd out private insurers. Sastry et al. (2025b) show that the presence of state guaranty funds increases incentives for insurers to exit the market.

period house prices and borrowing capacity upon disaster shocks when the collateral constraint becomes binding. Similarly, households do not internalize how high insurance demand today affects future insurance availability after disaster shocks when capital constraints become binding.

A social planner who faces the same constraints as the household internalizes these externalities and optimally manages both of the occasionally binding constraints. I decentralize the optimal policy with a tax on insurance, which can be either positive or negative. The tax varies with two distinct characteristics of the homeowners insurance market. First, the optimal subsidy or tax on insurance depends on how insurance protects aggregate house prices in the event of a disaster. Individual households ignore their impact on equilibrium house prices when disasters strike and when insurance coverage is low, while the planner internalizes this fire sale dynamic, justifying a subsidy on insurance.

Second, the optimal insurance subsidy or tax depends on the social value of insurer net worth in the economy. Households value insurers' net worth as it coincides with high insurance quantities. Individual households ignore their role in aggregate insurer net worth dynamics, while the planner uses insurance to balance recapitalization against crash-risk. Increasing insurance prices recapitalizes the insurer and increases its future risk-bearing capacity, which is beneficial for households. The planner can thus stimulate demand to fund the insurer and this motive becomes stronger when the marginal value of future insurers' net worth is high. Conversely, high insurance prices lead to high equilibrium insurance take-up, and make the insurer operationally leveraged. When highly leveraged, future supply reductions upon a disaster shock are steep. Overinsurance in good times can thus lead to steep contractions in bad times when weather risks materialize. I refer to this dynamic as the crash-risk channel. The planner balances the recapitalization and crash-risk channels in setting insurance prices.

To quantify the crash-risk channel, I introduce the concept of insurers' marginal propensity to insure (MPI), which characterizes insurers' risk-bearing capacity. MPI captures the sensitivity of the quantity of insurance policies sold to changes in insurer net worth. The concept thus reflects how well incumbent insurers are able to absorb disaster shocks and continue to supply insurance. MPI depends on capital constraints and disaster risk, but also on insurer investment returns, which act as a substitute for the insurer's product portfolio. When MPI is low, future insurance supply responds little to net worth shocks. The economy-wide MPI is an important concept in intertemporal trade-offs in insurance markets: insuring disaster risks provides macroeconomic benefits in the short run, but if the

MPI curve is sufficiently steep, high insurance take-up today could jeopardize the long run availability of insurance through the financial constraints of the insurer, justifying a tax on insurance.

In the calibrated model, the price-taking planner's insurance choice is higher than that of the household in the decentralized equilibrium and the difference becomes larger for low levels of savings (high levels of debt). When more indebted, the possibility of experiencing future binding collateral constraints is high. The value of insurance increases in these states as the planner internalizes possible house price crashes that lead to severe consumption drops. At high levels of savings, households self-insure and incentivizing insurance take-up is not necessary as the probability of reaching binding future constraints is zero. When insurer net worth is low and household savings high, the planner may even tax insurance slightly to prevent a non-linear decline in net worth upon a disaster.

The average insurance subsidy equals 2.7% and the planner's economy generates welfare gains equal to 0.2% of consumption equivalents relative to the decentralized equilibrium. The social planner achieves a lower insurance gap by demanding more. While the planner exposes insurers more to disaster risk, insurer fragility, as measured by MPI, is lower. The planner thus achieves a higher insurance penetration while avoiding insurance market crashes by appropriately subsidizing insurance across net worth states.

The insurance choice of the price-setting planner is lower than in the DE. When the insurer's supply curve is sufficiently steep, small changes in insurance quantities translate into large changes in insurance prices. When the reduction in marginal cost of insurance exceeds the reduction in marginal benefit of future insurer net worth, the price-setting planner has an incentive to continue reducing prices and de-fund the insurer. Subsequently, when equilibrium insurance prices fall, the attractiveness for insurers to operate in the market reduces. As available new worth lowers, insurer financial constraints are amplified, and the incentive for the planner to lower insurance prices grows further, so that low insurance take-up becomes persistent. This paper overall offers two policy implications: (i) while a subsidy on insurance is optimal in most states, its magnitude decreases when high insurance demand coincides with a high probability of future insurer net worth collapses, (ii) a regulator influencing insurance prices directly reduces welfare as it generates persistent under-insurance.

Empirically, I provide new motivating evidence on how insurance contractions transmit into housing markets. First, I document that insurer financial constraints have contributed to insurance contractions. Contractions in the zip codes with highest fire risk in California were

especially steep when the zip code was served by insurers with low capital ratios. Second, instrumenting insurance prices and non-renewals with out-of-zip-code disaster damages, I find that, conditional on weather risk, insurance price increases and non-renewals negatively affect house prices. Third, disaster shocks lower house prices in the struck regions especially when insurance coverage is low and household indebtedness is high, a dynamic that the theoretical model replicates.

This paper makes three contributions. First, I contribute to the literature studying the effects of insurance market regulation in homeowners- and capital markets. Capital regulation can distort insurance pricing and supply (Koijen and Yogo, 2015), affect insurers' appetite for insuring natural disaster risk (Born and Klimaszewski-Blettner, 2013), and affect insurers' risk-taking (Becker and Opp, 2013). Insurance rate caps can drive insurer exits (Oh et al., 2021, Liu and Liu, 2024). While the literature estimates regulatory distortions in reduced form, structural work is needed to understand which conditions justify insurance regulatory intervention and which regulation is optimal in disaster insurance markets. My main contribution is deriving the optimal time-consistent insurance policy in a dynamic housing model calibrated to the California housing market and informing state-dependent insurance regulation that jointly manages household collateral and insurer capital constraints.

Second, my paper contributes to the literature that investigates demand- and supply-side frictions as drivers of equilibrium insurance quantities. On the demand side, financial constraints (Sastry et al., 2025a) and behavioral explanations such as tail-risk neglect (Kunreuther and Pauly, 2004), learning (Gallagher, 2014) and limited attention (Cookson et al., 2024) explain insurance take-up. On the supply side, institutional constraints (Koijen and Yogo, 2022), regulatory frictions (Oh et al., 2021), informational frictions (Boomhower et al., 2024), capital market frictions (Eastman et al., 2024, Solomon, 2024) and coordination failures between insurers (Ibragimov et al., 2009) affect insurance supply. Evaluating the equilibrium effects of regulatory interventions comprehensively requires modeling how they affect insurance demand and supply simultaneously. To this end, I develop a dynamic model with occasionally binding constraints on both the household and intermediary side.

Third, I contribute to the literature studying how financial intermediaries shape macroeconomic fluctuations. The key insight in this literature is that financial frictions among intermediaries can amplify aggregate economic fluctuations. So far, the literature connecting intermediaries to business cycle fluctuations has focused on banks (Gertler and Kiyotaki, 2010, Coimbra and Rey, 2024, Favara and Imbs, 2015, Jamilov and Monacelli, 2025). Because insurance markets have first-order effects on housing markets, which are closely connected

to real economic activity, it is important to study how insurers affect the broader economy.[2] Property insurers furthermore fulfill a critical role in society by allowing households and firms to offload risks, potentially affecting consumption and investment through precautionary motives, and enabling financial contracts, such as mortgages and firm loans. This paper develops a dynamic stochastic general equilibrium model in which insurance markets, housing markets, and aggregate consumption co-move, which allows for understanding how insurance market conditions affect the real economy.

# 2 Motivating Empirical Evidence

I start by providing new empirical facts on contractions in weather-prone insurance markets. I analyze how insurer financing constraints affect the steepness of insurance market withdrawals, how insurance prices and policy cancellations affect house prices conditional on weather risk and how the interaction between insurance coverage and household leverage affect house price dynamics after disasters. Quantifying these effects reveal the cost of insurance contractions to homeowners and provide useful empirical benchmarks for the dynamic model I develop in section 3.

**Fact 1: The steepness of insurance market contractions increases with insurer financial constraints.**

I begin by showing trends in two insurance markets highly exposed to climate risk: California and Florida. Each year, I sort Floridan counties and Californian zip-codes into quintiles based on their hurricane and fire risk score, respectively.[3] Subsequently, within each hurricane or fire risk quintile, I sort regions into two groups based on the incumbent insurers' average risk-based capital ratios (RBC), weighted by their market shares. The county and zip-code average insurers' RBC ratios are based on the entire sample period and the sorts are thus time-invariant. The double sort allows me to assess the relevance of insurer financial conditions in the steepness of insurance contractions after conditioning on risk exposure. Figure 1 shows the time trend in policies issued by private insurers, the

[2] Property insurers in the U.S. provide over $15 trillion in multi-peril insurance coverage annually to homeowners.

[3] County-level hurricane risk scores are time-invariant, while zip-code fire risk scores vary over time. To standardize the sorting procedure, I average fire risk scores by zip-code. Averaging risk scores ensures that the composition of sorts does not change, preventing sudden changes in trends that result from changing portfolio compositions rather than market conditions.

market share of the residual insurer and average insurance premiums in both California and Florida for the each double sort.

The left panels reveal a sharp decline in privately issued insurance policies relative to the benchmark year 2009. While private market participation in low-risk segments in California has remained relatively stable or increased, the number of privately insured homeowners reduced sharply in both low- and high-risk segments. Contractions in the regions with highest weather risk were especially steep when the region was occupied by insurers with low capital ratios, which suggests that financial constraints playing a role in the steepness of insurance contractions. Even in low risk regions, growth was less or contractions were steeper when the region was served primarily by insurers with low capital ratios. Strikingly, even Floridan counties with low risk scores saw major exits, suggesting relatively low risk counties are also deemed to risky to insure.

The center panels indicate that conditional on risk exposure, the market share of the residual insurer is high when the region is occupied by insurers with low RBC ratio. This pattern is consistent across both regions and suggests that insurers financial constraints have contributed to the growth of residual markets. For example, in 2022, the share of the residual insurer FAIR plan in California in zip-codes with the highest fire risk was 40% when the zip-code was occupied by insurers with high capital ratios and 60% when occupied by insurers with low capital ratios.

The right panels show that homeowners insurance premiums have been rising sharply, primarily in high-risk segments. In Florida, average insurance premiums increase in insurer financial constraints. In California, in contrast, conditional on the region's weather risk, average insurance premiums decrease with insurer financial constraints. If insurer financial constraints correlate negatively with informational advantages, this is consistent with less-informed insurers underpricing informed insurers as in Boomhower et al. (2024).

Table A.4 formally tests the role of insurer financial conditions in increasing insurance premia and non-renewals in California. Columns 1-4 regress the change in zip-code level average insurance premia on changes on average insurer net worth and RBC ratios. Columns 5-8 perform the same regression with the log of non-renewals as the dependent variable. Instrumenting changes in insurer net worth and capital ratios with out-of-zip-code disaster losses reveals that financial constraints increase insurance contractions and drive rising insurance premia.

**Fact 2: Insurance market contractions negatively affect house prices conditional on weather exposure.**

**Homeowners Insurance Market Contractions by Risk and Insurer Financial Conditions**

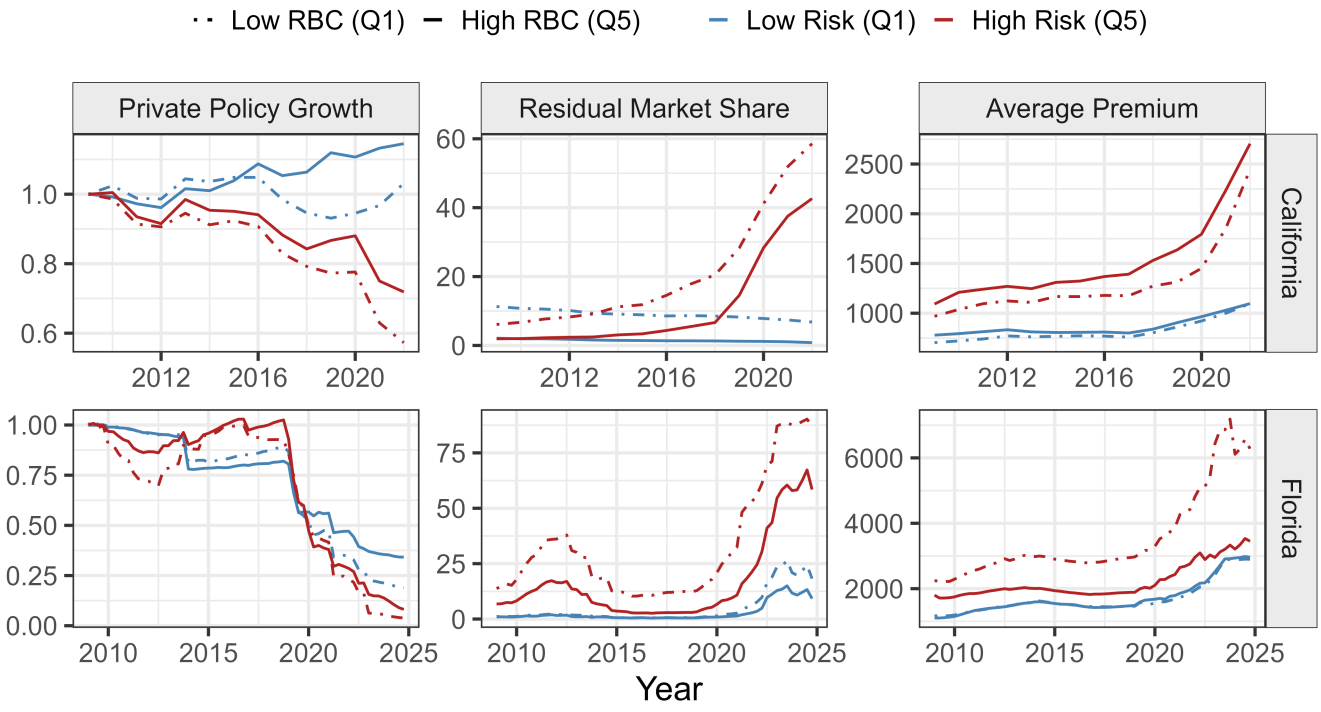



**Figure 1.** Insurance market trends in California (top row) and Florida (bottom row) in areas with differing disaster risk exposure. California (Florida) data is at the zip-code (county) level. Blue and red lines denote regions in the lowest (Q1) and highest (Q5) disaster risk quintile, respectively. Dash-dotted (solid) lines denote regions in which incumbent insurers have exposure-weighted average capital ratios below (above) the median. Private Policy Growth is the policy count from private insurers divided by the policy count in 2009. Residual Market Share captures the share of policies sold by the state residual insurer. Average Premium captures the nominal premium for a homeowners insurance policy that is renewed annually.

To illustrate the economic relevance of under-insurance beyond a shift in risk exposure from insurers to households, I investigate how insurance contractions transmit into housing markets. To test how insurance market conditions affect housing outcomes, I run the following specification:

$$Y_{z,t} = \delta M_{z,t} + \gamma' X_{c,t} + \eta_z + \iota_t + \epsilon_{z,t}, \tag{1}$$

where $Y_{z,t}$ is a zip-code level economic indicator, including the natural logarithm of median or mean house prices. The variable $M_{z,t}$ captures local insurance market conditions. I consider two proxies for $M_{z,t}$: (i) the logarithm of average homeowner insurance prices and (ii) the logarithm of the number of insurer cancellations. I construct insurance prices in zip code $z$

in year $t$ as the weighted average insurance prices over insurers active in zip code $z$:

$$\text{Insurance price}_{z,t} = \sum_{i \in I} w_{i,z,t} \times \frac{\text{Premiums}_{i,z,t}}{\#\text{Annual exposures}_{i,z,t}} \tag{2}$$

where

$$w_{i,z,t} = \frac{\text{Premiums}_{i,z,t}}{\sum_{i \in I} \text{Premiums}_{i,z,t}}$$

denotes the market share insurer $i$ in zip code $z$ in year $t$. The insurance price captures the average annual cost of a homeowners' insurance policy. I construct the number of insurer cancellation by summing the cancellations across insurers in given zip-code. These cancellations are initiated by the insurer and can thus be considered as a supply-driven change to insurance penetration.

The zip-code level fixed-effects $\eta_z$ accounts for time invariant unobserved heterogeneity in insurance conditions and economic outcomes, such as wildfire risk perceptions that could correlate with both insurance prices and house prices. Time fixed effect $\iota_t$ captures common shocks across zip-codes. The vector $X_{c,t}$ contains control variables. Following the literature (Eastman et al., 2024), I control for various time-varying economic and demographic characteristics that affect jointly affect house demand and supply, and insurance prices.

The set of county-level controls includes demographic attributes (population, population density, unemployment rate, college attainment, the proportion of Black and Hispanic populations, the percentage of people with health insurance), income attributes (median household income, the percentage of households below the poverty line), and housing market attributes (median housing value, median year of construction, the percentage of owner-occupied homes, the vacancy rate, and the percentage of owner-occupied homes with a mortgage). In addition, I control for disaster damages in the past 5 years.

The set of zip-code level controls include the average amount of fire coverage per policy, separately including the entire policy limit and the limit for the house only, the number of existing policies, the average deductible per policy and the logarithm of county-level disaster damages, the number of insurers, and zip-code level insurer market concentration measured by the Herfindahl index. These insurance variables account for households endogenous selection of contract features that correlate with insurance prices. In addition, the insurance market indicators account for correlations between growing weather risk, market power and insurance pricing.

The ideal experiment to estimate the effects of insurance contractions on the economy would capture changes in insurers' supply decisions that are unrelated to risk exposure

and local insurance demand. A key empirical challenge in understanding how insurance contractions affect macro-economic outcomes is that insurance prices simultaneously reflect climate risk exposure, the cost of living and possibly historical damages that directly affected property values. Proxies for insurance contraction, including insurance prices and policy cancellations, could affect macro-economic outcomes through each of these channels, while I am primarily interested in understanding the effects of insurance contractions conditional on disaster exposure and disaster realizations. While one could correct for observable disaster damages and risk scores, non-linear relationships between disasters, insurance payouts and property values could lead to model misspecification. In addition, equilibrium insurance prices are driven by both supply and demand. Potentially, high insurance premia reflect historical high insurance demand through increased awareness about climate risk that correlates indirectly with households wealth, leading to biased estimates.

I design an empirical strategy that isolates insurer supply shocks by exploiting the geographical breakdown of insurers' activities. I instrument average zip-code level insurance prices and policy cancellations by disaster shocks occurring in other counties weighted by the incumbent insurers' out-of-zip-code exposure shares. The empirical strategy is based on the testable assumption that, disaster losses in one region that drive the insurer to become financially constrained or adopt a conservative pricing strategy could affect insurance prices and supply decisions in other regions as well. Section 9.1 in the Appendix describes the construction of this instrument in more detail and shows the predictive power in the first-stage.

Table 1 shows the second stage results of the regression of house prices on insurance indicators in both the OLS and IV specifications. The OLS specification shows that a 10% increase in insurance prices lowers median and average house prices by 0.8-1%, respectively. For average hosue prices, the IV specifications yield similar coefficients. Column 3 and 4 show that insurer non-renewals lower house prices. Finally, table A.5 shows that increases in the market share of the residual insurer similarly lower house prices. Overall, these results suggest that insurance contractions are costly for homeowners given their negative effect on house prices.

**Fact 3: The interaction of insurance and debt pins down house price fluctuations upon weather shocks.**

The results thus far show that insurance prices and contractions can affect house prices conditional on disaster risk. Insurance conditions can also affect house price indirectly through disaster realizations. In the event of severe weather impacts, insurance can protect collateral

**Table 1**
Regression of house prices on insurance indicators in OLS- and IV specification. In the IV specifications, insurance prices are instrumented using disaster shocks occurring in other counties weighted by the incumbent insurers' out-of-zip-code exposure shares. Non-renewals are instrumented by the logarithm of out of zip-code weather losses weighted by insurers' portfolio weights and local market shares. Standard errors in parentheses are clustered at the zip-code level. Significance codes: ***: 0.01, **: 0.05, *: 0.1

| | Log(Median HP) | | Log(Mean HP) | | Log(Median HP) | | Log(Mean HP) | |
|---|---|---|---|---|---|---|---|---|
| | OLS | IV | OLS | IV | OLS | IV | OLS | IV |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Log Ins. Price | $-0.08^{***}$ | 0.05 | $-0.10^{***}$ | $-0.08^{**}$ | | | | |
| | (0.03) | (0.05) | (0.02) | (0.03) | | | | |
| Log Non-renewals | | | | | $-0.02^{***}$ | $-0.08^{**}$ | $-0.002$ | $-0.05^{***}$ |
| | | | | | (0.006) | (0.04) | (0.003) | (0.02) |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Zip-code FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 12,768 | 12,731 | 11,753 | 11,746 | 7,361 | 7,361 | 7,002 | 7,002 |
| $R^2$ | 0.96 | 0.96 | 0.99 | 0.99 | 0.98 | 0.98 | 0.99 | 0.99 |
| Within $R^2$ | 0.08 | 0.07 | 0.16 | 0.15 | 0.04 | 0.04 | 0.17 | 0.17 |

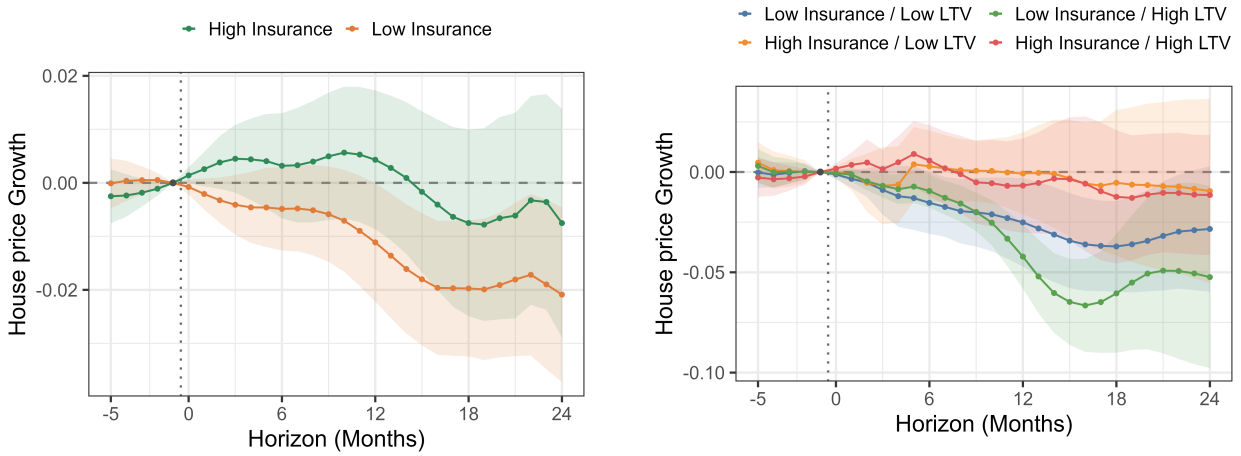
values by enabling households to reconstruct their homes. Insurance could prevent defaulting on ones mortgage and mitigate reductions in house price when households otherwise need to sell leftover structures at discounts. In contrast, high insurance coverage could facilitate moving rather than rebuilding and lower the supply of housing after disaster impacts, increasing house prices (Zivin et al., 2023). To test how insurance coverage interact with disaster realization and transmit into house prices, I run the following local projection:

$$Y_{z,t+h} - Y_{z,t-1} = \delta_h \cdot \mathbf{1}_{z,t}^{fire} \cdot \text{High Insurance}_{z,t} + \gamma' X_{c,t} + \eta_z + \iota_m + \varepsilon_{i,t+h} \tag{3}$$

where $\mathbf{1}^{fire}$ takes on value 1 if the zip-code experienced a severe wildfire and $\iota_m$ are year-month fixed effects. I define severe wildfires based on whether the zip-code is within or adjacent to a fire perimeter after a declared state of emergency is issued by the Governor of the state California. This data is available from 2019 onwards. $\text{High Insurance}_{z,t}$ equals one if the average amount of fire coverage per policy is higher than the median in a given year and zero otherwise. I include $h \in \{0, 1, \ldots, 6\}$ in the estimation. As before, $Y_{z,t}$ is the natural logarithm of median or mean house prices. I use the same set of controls as in regression (1) except I include 2 additional lags of the natural logarithm of median and mean

house prices. I compute the response of house prices following a severe wildfire separately for the low- and high insurance group. Standard errors are clustered by zip-code.

Figure 2 shows that following wildfire incidents, house prices decrease significantly in zip-codes in which households have low fire insurance coverage. Specifically, average house prices drop by 3% by year 2 in zip-code that have the least fire coverage relative. In contrast, the house price drop in zip codes with high insurance coverage for following wildfire incidents is significantly less steep and recover more quickly (Panel a). House price reductions are especially steep if low insurance is paired with high LTV ratios (Panel b), reaching 8% after 3 years. These heterogeneous effects show that the interaction between insurance and debt plays an important role in how disaster shocks affect house prices. I do not find these effects for median house prices, which is consistent with the effects of wildfires on house prices concentrating among expensive houses.




**(a) By Insurance Coverage**

**(b) By Insurance Coverage and Indebtedness**

**Figure 2.** Percentage change in average zip-code house prices following wildfire incidents, by insurance coverage (panel a) and by insurance coverage and household indebtedness (panel b). Zip-codes are sorted into two insurance coverage groups based on whether their average fire coverage per policy is above or below the median. In panel (b), zip-codes within each insurance coverage group are further sorted into two indebtedness groups based on whether households' average loan-to-value ratio is above or below the median. Horizons are in months relative to the wildfire. Shaded areas reflect 95% confidence intervals. Standard errors are clustered at the zip-code level.

# 3 A dynamic housing model with insurance intermediaries

The empirical results show that insurance contractions are costly for homeowners by lowering house prices conditional on weather risk. Low insurance in combination with high LTV ratios furthermore drives sharp house price reductions following disasters. I develop a model of a small-open economy to study the interactions between insurance and housing markets in the presence of disaster risk. I first investigate how disaster shocks propagate to housing markets through insurance intermediaries. Second, I characterize the optimal insurance allocation that internalizes insurer fragility and the benefits of insurance to homeowners. Third, I compute the implicit subsidy or tax on insurance that generates this allocation.[4]

## 3.1 Disaster risk

Time is discrete and infinite. Disasters occur with probability $\pi^d$. Denote $\mathbf{1}_t$ as the indicator that equals 1 if a disaster occurred and 0 otherwise. Upon impact, a disaster destroys houses by a fraction $\phi$, which I refer to as disaster severity. Weather risk $\theta_t$ is time-varying and follows an auto-regressive process:

$$\theta_t = \rho_\theta \theta_{t-1} + \varepsilon_{\theta,t} \quad \varepsilon_{\theta,t} \sim \mathcal{N}(0, \sigma_\theta^2) \tag{4}$$

I set $\phi = \frac{1}{1+\exp(-\theta)}$. The logistic transformation ensures that the fraction of destroyed houses is between 0 and 1. I discretize the process of weather risk using the Tauchen method.

## 3.2 Households

Households derive utility from consumption $c_t$, and housing $s_t$. Households can invest $b_{t+1}$ into bonds to smooth consumption. Households invest in housing $s_{t+1}$, purchase insurance $x_{t+1}$ to protect their housing assets and redraw home equity $l_t$. The representative household maximizes:

$$\max_{\{c_t, b_{t+1}, l_t, x_{t+1}, s_{t+1}\}} \mathbb{E}_t \sum_{k=0}^{\infty} \beta^k u(c_t, s_t) \tag{5}$$

[4] The model is a variation of the frameworks proposed by Bianchi and Mendoza (2018) and Jamilov and Monacelli (2025). I modify Bianchi and Mendoza (2018)'s model by incorporating housing rather than capital that is subject to disaster shocks, and allowing households to insure themselves in a private insurance market. I modify Jamilov and Monacelli (2025) by incorporating an intermediary in the form of an insurer rather than a bank.

where the utility takes the following form.

$$u(c_t, s_t) = \frac{1}{1-\psi} \left(c_t + \nu_s g(s_t)\right)^{1-\psi} \tag{6}$$

The utility function features constant relative risk aversion (CRRA) over consumption and housing and is similar to the utility specification of Iacoviello (2005). Households choose next-period housing and current housing enters as a state variable in the household problem. Denote by $\psi$ the coefficient of relative risk aversion. The parameter $\nu_s$ captures the utility weight on housing. The function $g(s_t) = \frac{s_t^{1-\sigma_s}}{1-\sigma_s}$ is increasing and concave in $s_t$. Utility is intertemporally separable but not intratemporally separable. Finally, the utility function satisfies the Inada conditions. The budget constraint reads

$$c_t + \frac{b_{t+1}}{R_t} + s_{t+1} p_t^s + p_t^x x_{t+1} s_{t+1} = b_t + w_t + s_t p_t^s - \mathbf{1}_t \phi_t p^r s_t + x_t \mathbf{1}_t \phi_t p^r s_t + l_t - f(l_t) \tag{7}$$

where $b_t$ denotes the stock of one-period bonds issued in the last period. $R_t$ is the exogenous non state contingent world interest rate that satisfies $\beta R < 1$. Labor is normalized to 1 so that total labor income is $w_t$. The interest rate is stochastic and follows a stationary Markov process similar to disaster severity. While Jamilov and Monacelli (2025) endogenize the risk-free rate inside their banking model, I follow Bianchi and Mendoza (2018) by assuming that the economy is small and open relative to world financial markets. Similarly, I assume that participating insurers have foreign owners for simplicity, which rules out that insurance demand is driven by the possibility to recollect premiums through future dividend streams.

Housing $s_{t+1}$ can be purchased at market price $p_t^s$. Housing assets are in fixed supply, so the market clearing condition is simply $s_t = 1$. Finding an equilibrium in the housing market involves finding a $p_t^s$ such that, given household optimal decisions, the market clearing condition holds. The amount of housing purchased in the previous period that enters the budget constraint is $s_t$. I assume that following a disaster households rebuild immediately and incur rebuilding cost $\mathbf{1}_t \phi_t p^r s_t$, where $p^r$ captures the cost to rebuild a house of size 1. The total supply of housing thus remains constant. The rebuilding cost $p^r$ is assumed to be constant for simplicity.

Households can buy insurance $x_{t+1}$ against next-period disaster risk from insurers. Insurance coverage is defined as the fraction of housing that is potentially destroyed in the next-period and can be bought at price $p_t^x$ per unit of the exposed house.[5] The insurance

[5] Defining insurance cover as the fraction of possible future housing damage avoids modeling insurance payouts conditional on exceeding the insurance contract limit, which would preclude the differentiability of the problem.

quantity bought gets aggregated across all insurers $i$, i.e. $x_t = \int_0^1 x_t(i)di$. The insurance payout based on insurance purchased in the previous period is conditional on a disaster having occurred and equals $x_t \mathbf{1}_t \phi_t p^r s_t$. The insurance payout is thus proportional to the replacement value and not the market value of the house, which is standard in insurance contracts. The total insurance premium to cover next-period losses equals $p_t^x x_{t+1} s_{t+1}$ and is thus proportional to the housing investment, the fraction of the possible housing damage the household wishes to cover, the rebuilding cost $p^r$ and the price per unit of insurance.

Households can redraw equity from their housing investments $l_t$, which comes at a cost $f(l_t)$ that captures transaction costs. The function $f(l_t)$ is strictly increasing in $l_t$, strictly convex and positive. Home equity redraws are small and play a modest role in the dynamics of the economy, but aid in the identification of the shadow value on the borrowing constraint.

**Borrowing constraint** The household can obtain an intertemporal loan $-b_{t+1}/R_t$ that involves interest payments. Borrowed funds can be diverted, which is avoided by imposing a collateral constraint. The collateral constraint limits total debt not to exceed a fraction $\kappa_t$ of the market-value of beginning-of-period housing assets, which equals the post-disaster housing plus the insurance payout minus the home equity redraw. The leverage constraint can thus be written as:

$$-\frac{b_{t+1}}{R_t} \leq \kappa_t \left( p_t^s s_t \left(1 - \mathbf{1}_t \phi_t\right) + x_t \mathbf{1}_t \phi_t p^r s_t - l_t \right) \tag{8}$$

The left-hand side equals total borrowing. The right-hand side equals housing collateral plus insurance payouts after the realization of any disaster shocks minus home equity redraws. Hence, $\kappa_t$ puts a ceiling on the leverage ratio. This setup implies that disaster damages and insurance payouts affect collateral values and new borrowing capacity. Home equity redraws, by taking away collateral, further tighten the borrowing constraint.

**Optimality conditions** The representative household chooses $\{c_t, b_{t+1}, l_t, x_{t+1}, s_{t+1}\}$ to maximize their expected lifetime utility (5) subject to the budget constraint (7) and the borrowing constraint (8). In the baseline model, the household takes prices and insurer net worth as given. This maximization yields the following optimality conditions and constraints at each date $t$ (see Appendix 10):

$$c_t :: \qquad U'(t) = \lambda_t \tag{9}$$

$$b_{t+1} :: \qquad U'(t) = \beta R_t \mathbb{E}_t \left[ U'(t+1) \right] + \mu_t \tag{10}$$

$$l_t :: \qquad 1 = f'(l_t) + \frac{\mu_t \kappa_t}{\lambda_t} \tag{11}$$

$$x_{t+1} :: \qquad p_t^x = \mathbb{E}_t \left[ (m_{t+1} + \frac{\mu_{t+1}\kappa_{t+1}}{\lambda_t}) \mathbf{1}_{t+1} \phi_{t+1} p^r \right] \tag{12}$$

$$s_{t+1} :: \qquad p_t^s = \mathbb{E}_t \left[ m_{t+1} \left\{ \nu_s g'(s_{t+1}) + p_{t+1}^s \right\} + \frac{\mu_{t+1}\kappa_{t+1}}{\lambda_t} Q_{t+1} \right] \tag{13}$$

$$c_t + \frac{b_{t+1}}{R_t} + s_{t+1} p_t^s + p_t^x x_{t+1} s_{t+1} = b_t + w_t + s_t p_t^s + (x_t - 1)\mathbf{1}_t \phi_t p^r + l_t - f(l_t) \tag{14}$$

where $Q_{t+1} = p_{t+1}^s (1 - \mathbf{1}_{t+1}\phi_{t+1})$ is the post-disaster value of the house, $\lambda_t = \tilde{C}_t^{-\psi} = U'(t)$ is the marginal utility of consumption given the composite $\tilde{C}_t = c_t + \nu_s g(s_t)$ and $m_{t+1} = \beta \lambda_{t+1}/\lambda_t$ is the stochastic discount factor.

Equation (9) is the first-order of the Lagrangian with respect to consumption. Equation (10) is the Euler equation with $\mu_t$ denoting the multiplier on the borrowing constraint. It states that the marginal benefit of borrowing exceeds the marginal cost by the value of relaxing the borrowing constraint $\mu_t$. Condition (11) states that the marginal benefit of redrawing home equity should be equal to its marginal cost plus a financing cost when the collateral constraint becomes binding. The state contingent insurance price in equation (12) captures the cost to cover the expected damage. The insurance price equation sets the demand for insurance by equalizing the marginal benefit and the marginal cost. The marginal benefit equals the expected next-period's damage, adjusted for expected collateral constraints, multiplied by the stochastic discount factor $m_{t+1}$. If disasters lead to binding constraints in the next-period in expectation, households are willing to pay more for insurance. A binding collateral constraint implies selling houses at fire sale prices to repay debt, which is costly. The fact that insurance helps prevent hitting binding collateral constraints in the next period is why the expectation over $\mu_{t+1}$ ends up in the insurance valuation. The insurance price is furthermore convex in the disaster severity state $\theta$. A high disaster severity today, given the AR(1) process, leads to allocating more weight to high disaster severity states in the future as well, increasing the expected damage and therefore the insurance price.

The housing condition (13) equates the marginal cost of investing in housing today with the marginal benefit of housing in the next period. High insurance premia today may reduce house prices by affecting the marginal utility of consumption. The marginal benefit is the sum of the utility gain and capital gains, with the latter being the housing investment valued at next-period price $p_{t+1}^s$ adjusted for any disaster damages, and insur-

ance payouts. Whenever $\mu_{t+1} > 0$ the collateral constraint binds in the next period and the fact that post-disaster housing serves as collateral increases the marginal benefit by $\kappa_{t+1}\left(p^s_{t+1}(1-\mathbf{1}_{t+1}\phi_{t+1}) + x_t\mathbf{1}_t\phi_t p^r s_t - l_t\right)$.

One can define the gross return on housing assets as:

$$R^s_{t+1} = \frac{\nu_s g'(s_{t+1}) + p^s_{t+1}}{p^s_t} \tag{15}$$

which states that the return on housing assets represents the marginal utility of housing plus the value of the house tomorrow including insurance payouts divided by the investment into the house today including insurance costs. Housing utility affecting house prices is consistent with the large literature studying house prices as a function of housing amenities through hedonic regressions. If the collateral constraint does not bind, i.e., $\mu_t = 0$, then iterating on (13) and using the transversality condition, we can express the house price as the expected present value of housing amenities discounted with $R^s_{t+1}$ (as shown in appendix 10):

$$p^s_t = \mathbb{E}_t\left[\sum_{j=0}^{\infty}\left(\prod_{i=0}^{j}\mathbb{E}_{t+i}R^s_{t+1+i}\right)^{-1}\nu_s g'(s_{t+j+1})\right] \tag{16}$$

In a world without financing constraints, the value of the house net of insurance payments is given solely by the discounted future housing benefits. We can combine equations (10) and (13) to derive the expected excess return on housing $R^{hp} \equiv \mathbb{E}_t\left[R^s_{t+1} - R_t\right]$:

$$R^{hp}_t = \mathbb{E}_t\left[R^s_{t+1} - R_t\right] = \underbrace{\frac{\mu_t}{\lambda_t\,\mathbb{E}_t\left[m_{t+1}\right]}}_{\text{Liquidity premium}} - \underbrace{\frac{\mathbb{E}_t\left[m_{t+1}\,\Upsilon_{t+1}\right]}{\mathbb{E}_t\left[m_{t+1}\right]}}_{\text{Collateral effect}} - \underbrace{\frac{\text{Cov}_t\left(m_{t+1}, R^s_{t+1}\right)}{\mathbb{E}_t\left[m_{t+1}\right]}}_{\text{Risk premium}} \tag{17}$$

where

$$\Upsilon_{t+1} = \frac{\kappa_{t+1}\mu_{t+1}p^s_{t+1}}{\lambda_{t+1}p^s_t} \tag{18}$$

is the marginal value of relaxing the borrowing constraint at time $t+1$. The liquidity premium increases the return on housing when the collateral constraint binds as housing assets are illiquid and therefore cannot be used for consumption. The collateral effect partially offsets the liquidity premium. Investing in housing eases collateral constraints tomorrow, which reduces possible fire sales, and therefore affects expected excess returns through $\Upsilon_{t+1}$. Finally, households demand a risk premium when investing in housing. The covariance term tends to be negative, that is, low housing returns coincide with states where marginal utility of consumption is high.

If the liquidity premium dominates, then equations (40) and (17) imply a fire-sale mechanism. When collateral constraints become binding, the expected excess return on housing increases, and house prices today decrease. This can be interpreted as households collectively needing to sell assets in order to repay debt, which lowers equilibrium house prices. This in turn lowers aggregate borrowing capacity in the economy. The binding collateral constraint can result from either over-borrowing as in Bianchi and Mendoza (2018) or under-insurance that leads to more collateral being destroyed upon a disaster.

Overall, insurance take-up and affordability affect collateral values in two ways. First, low insurance take-up can result in a higher share of households becoming credit constrained upon disaster impact as they don't receive disaster payouts through (7). Low insurance payouts tighten the leverage constraint upon a disaster realization and drive housing fire sales tomorrow. Second, high insurance premia lower house price today through (13) and could make the leverage constraint bind today.

### 3.3 Dynamic household problem

Denote $\mathcal{S}_t = \{\mathbf{1}_t, \theta_t, R_t\}$ as the realization of shocks at date $t$. The realization of a disaster $\mathbf{1}_t$ even though it is i.i.d, is a state variable as it is needed to compute of today's wealth and housing collateral.[6] The household state variables are the individual states $(s_t, b_t, x_t)$ and the aggregate states $(B_t, X_t, n_t, \mathcal{S}_t)$. The two endogenous state variables for the household are borrowing $B_t$ and insurance $X_t$ as they both affect future collateral and therefore future house prices. To form expectations about future house prices, households have a conjectured pricing function $p^s(B_t, X_t, n_t, \mathcal{S}_t)$, $p^x(B_t, X_t, n_t, \mathcal{S}_t)$ and $w(B_t, X_t, n_t, \mathcal{S}_t)$, and a perceived law of motion for the aggregate bond position $B_{t+1} = \Gamma^B(B_t, X_t, n_t, \mathcal{S}_t)$ and aggregate insurance position $X_{t+1} = \Gamma^X(B_t, X_t, n_t, \mathcal{S}_t)$. I need to track insurer net worth $n_t$ as insurance prices, and therefore equilibrium insurance take-up, vary with insurer financial conditions. Yet, from the household's perspective, insurer net worth is exogenous. Finally, aggregate housing stock is not a state variable given that it is in fixed supply. The household's recursive optimization problem is:

$$\begin{aligned} V(s_t, b_t, x_t, n_t, \mathcal{S}_t, B_t, X_t) = \max_{c_t, s_{t+1}, l_t, x_{t+1}, b_{t+1}} \{ & u(c_t, s_t) \\ & + \beta \mathbb{E}_{\mathcal{S}_{t+1}|\mathcal{S}_t} V(s_{t+1}, b_{t+1}, x_{t+1}, n_{t+1}, \mathcal{S}_{t+1}, B_{t+1}, X_{t+1}) \Big\} \end{aligned} \tag{19}$$

[6] Alternatively, one can track collateral values and drop $\mathbf{1}_t$ as a state-variable, but this would not change the number of state variables. Tracking collateral in addition requires more grid points, which goes at the expense of computational speed.

subject to

$$
\begin{aligned}
c_t + \frac{b_{t+1}}{R_t} + s_{t+1}p_t^s + p_t^x x_{t+1}s_{t+1} &= b_t + w_t + p_t^s s_t + \mathbf{1}_t \phi_t s_t x_t p^r + l_t - f(l_t) \\
-\frac{b_{t+1}}{R_t} &\leq \kappa_t \left(p_t^s s_t \left(1 - \mathbf{1}_t \phi_t\right) + \mathbf{1}_t \phi_t s_t x_t p^r - l_t\right) \\
B_{t+1} &= \Gamma^B(B_t, X_t, N_t, \mathcal{S}_t) \\
X_{t+1} &= \Gamma^X(B_t, X_t, N_t, \mathcal{S}_t) \\
N_{t+1} &= \Gamma^N(B_t, X_t, N_t, \mathcal{S}_t)
\end{aligned}
$$

The solution of this problem contains the policy rules $\hat{c}_t(b_t, x_t, n_t, s_t, B_t, X_t, \mathcal{S}_t)$, $\hat{l}_t(b_t, x_t, n_t, s_t, B_t, X_t, \mathcal{S}_t)$, $\hat{s}_{t+1}(b_t, x_t, n_t, s_t, B_t, X_t, \mathcal{S}_t)$, $\hat{x}_{t+1}(b_t, x_t, n_t, s_t, B_t, X_t, \mathcal{S}_t)$, and $\hat{b}_{t+1}(b_t, x_t, n_t, s_t, B_t, X_t, \mathcal{S}_t)$. The decision rules for bonds and insurance induce an actual law of motion for aggregate bonds, insurance, and net worth, given by $\hat{b}_{t+1}(\mathbf{1}, B_t, x_t, n_t, \mathcal{S}_t, B_t, X_t)$, $\hat{x}_{t+1}(\mathbf{1}, B_t, x_t, n_t, \mathcal{S}_t, B_t, X_t)$, and $\hat{n}_{t+1}(\mathbf{1}, B_t, x_t, n_t, \mathcal{S}_t, B_t, X_t)$, respectively. Finally, the equilibrium contains actual pricing functions $\hat{p}^s(n_t, \mathcal{S}_t, B_t, X_t)$, $\hat{p}^x(n_t, \mathcal{S}_t, B_t, X_t)$ and $\hat{w}(n_t, \mathcal{S}_t, B_t, X_t)$.

## 3.4 Insurers

Whether insurance is available at affordable prices depends on insurers' risk-bearing capacity. This section outlines how insurers supply disaster insurance in the economy. There is a continuum of insurers of measure one selling insurance $x_{t+1}$ to households and investing the proceeds into capital producing firms. The insurer is risk-neutral, accumulates net worth $n_t$, and maximizes the present discounted value of dividends, $J_t$. Insurers start with an initial net worth $n \in \mathcal{N} \subset \mathbb{R}_+$, and choose an asset and liability configuration after paying out any dividends. How much insurance insurers can sell is constrained by regulation.

At the start of the period, insurers decide on how much dividends to pay out, which are defined as the fraction $\delta_t$ of beginning of the period net worth. Retained net worth available for investment equals:

$$N_t = (1 - \delta_t) n_t \tag{20}$$

I assume that insurers face a non-pecuniary quadratic cost $f(\delta_t)$ over paying out dividends, which captures preferences of managers for dividend smoothing (Jermann and Quadrini, 2012). Whether insurers pay out dividends depends on the attractiveness of participating in the market. When insurer net worth is low, regulatory constraints may bind. In this case, the marginal value of net worth is high, reducing dividend payouts. Furthermore, when

insurer net worth is low, the aggregate stock of capital is low, raising expected asset returns and increasing the attractiveness of selling insurance. Equilibrium dividend payouts thus overall increase in net worth, providing a natural upper bound on the growth of insurer net worth.

After paying out dividends, insurers decide on how much insurance to supply taking insurance prices and expected asset returns as given. Insurers collect premiums from sold insurance that can be directly invested. Possible claims are settled at time $t+1$. The balance sheet constraint that must hold at all times is given by[7]:

$$a_t = N_t + p_t^x x_{t+1} s_{t+1} \tag{21}$$

Insurer assets that are invested thus equal net worth today plus insurance premiums. Let $R_t^k$ be the return on the insurers investment portfolio from period $t-1$ to period t. The law of motion of insurer-level net worth, $n_t$, can be written as:

$$n_t = a_{t-1}(1 + R_t^k) - \mathbf{1}_t \phi_t x_t p^r - \chi_1 x_t^{\chi_2}$$

Net worth today is the sum of prior period assets that generated return $1 + R_t^k$ minus claim payouts and operational costs. Claim payouts occur only when a disaster occurred and are proportional to the insurance exposure chosen in the previous period and costs to rebuild $p^r$. The introduction of convex operational costs allows for matching decreasing marginal propensities to insure across the insurer net worth distribution.

**Regulatory friction** Similar to Gertler and Kiyotaki (2010) and Jamilov and Monacelli (2025), I consider that the intermediary is subject to an incentive constraint. In the economy with insurers, the constraint limits insurers' ability to raise funding in the insurance market for investment purposes. The regulation constrains the insurer by ensuring that the insurer is able to pay out claims in most cases and thus acts as a value-at-risk constraint. Let $W_t(N_t; \theta_t)$ be the maximized value given investment and insurance decisions at the end of period $t$ after dividend payouts. The regulation imposes that the insurer value is at least a fraction $\zeta$ of its disaster exposure:

$$W(N_t; \theta_t) \geq \zeta x_{t+1} p^r s_{t+1} \tag{22}$$

[7] Selling insurance generates liabilities, but at policy inception premium income is booked as an unearned premium reserve, which directly offsets the premiums collected on the asset side. Selling insurance therefore generally has no contemporaneous effect on net worth.

The regulatory constraint thus puts a cap on the amount of insurance the insurer can sell. Since collected premiums are invested, the constraint therefore indirectly caps investable assets as well.

**Dynamic insurer problem** The relevant state variables for the insurer are net worth $n_t$ and time-varying disaster severity $\theta_t$. The dynamic problem for the insurer is:

$$\begin{aligned} J(n_t, \theta_t) &= \max_{\delta_t} \{\delta_t n_t - f(\delta_t) + W(N_t, \theta_t)\} \\ W(N_t, \theta_t) &= \max_{x_{t+1} \geq 0} \beta \, \mathbb{E}_{\theta_{t+1}|\theta_t} \left[ J(n_{t+1}, \theta_{t+1}) \right] \end{aligned} \tag{23}$$

subject to

$$\begin{aligned} N_t &= (1 - \delta_t) n_t \\ a_t &= N_t + p_t^x x_{t+1} s_{t+1} \\ n_t &= a_{t-1}(1 + R_t^k) - \mathbf{1}_t \phi_t x_t p^r - \chi_1 x_t^{\chi_2} \\ \zeta x_{t+1} p^r &\leq W(N_t, \theta_t) \end{aligned}$$

where $W(N_t, \theta_t)$ captures the value of underwriting after paying out dividends.

One can substitute out the balance sheet constraint and the law of motion for net worth to obtain a simplified problem:

$$W(N_t, \theta_t) = \max_{\{x_{t+1}\} \geq 0} \left\{ \mathbb{E}_t \, \mathcal{M} \Big[ \left( p_t^x (1 + R_{t+1}^k) - \mathbf{1}_{t+1} \phi_{t+1} p^r \right) x_{t+1} + N_t (1 + R_{t+1}^k) - \chi_1 x_{t+1}^{\chi_2} \Big] \right\} \tag{24}$$

where $\mathcal{M} = \beta \, \frac{J(n_{t+1}, \theta_{t+1})}{n_{t+1}}$ is the insurers' augmented discount factor.

**Unconstrained region** Taking the first-order condition of (24) with respect to $x_{t+1}$ when the regulatory constraint does not bind yields:

$$\mathbb{E}_t \Big[ \mathcal{M} \Big( p_t^x (1 + R_{t+1}^k) - \mathbf{1}_{t+1} \phi_{t+1} p^r - \chi_1 \chi_2 \Big( x_{t+1}^{uncon} \Big)^{\chi_2 - 1} \Big) \Big] = 0 \tag{25}$$

where $x_{t+1}^{uncon}$ is the optimal insurance supply of the insurer when the regulatory constraint is not binding. When the regulatory friction is not binding, insurers sell insurance up to the point where the marginal value of generating investable premiums equals the marginal cost of the insurance portfolio, consisting of expected claims and operational costs. When investments are profitable, insurers seek to sell more insurance that generates investable

premiums. This effect increases in the insurance price at which insurers can sell insurance. Even if premiums are below expected claim payouts, insurers choose to sell insurance if the premiums can be invested at a sufficiently high yield.

Define the marginal propensity to insure (MPI) as the sensitivity of insurance supply with respect to insurer net worth. In the unconstrained region, insurance supply does not depend on retained net worth, so that $MPI^{uncon} = \frac{\partial x_{t+1}}{\partial N_t} = 0$.[8]

**Constrained region** When the constraint binds, one can substitute $\zeta x_{t+1} p^r s_{t+1} = W(N_t, \theta_t)$ and derive the insurers' policy function for insurance $\tilde{x}_{t+1}$ as an implicit function of choice variable $x_{t+1}$:

$$\tilde{x}^{con}_{t+1}(n, \theta) = \frac{\mathbb{E}_t \left[ \mathcal{M} \left( n_t(1 + R^k_{t+1}) - \chi_1 x^{\chi_2}_{t+1} \right) \right]}{\zeta p^r - \mathbb{E}_t \left[ \mathcal{M} \left( p^x_t (1 + R^k_{t+1}) - \mathbf{1}_{t+1} \phi_{t+1} p^r \right) \right]} \tag{26}$$

As in the unconstrained region, insurance supply is increasing in expected asset returns and insurance prices, and decreasing in expected claims and operational costs. Insurance supply furthermore decreases in regulatory stringency $\zeta$. In addition, insurance supply depends on net worth as net worth increases the value function $W_t$ and therefore how severely capital constraints bind.

If constrained by regulation, insurers can only sell insurance up to the point where the regulatory constraint becomes binding, leaving a wedge between the marginal value of assets and the marginal cost of insurance. The spread between the return on the investment portfolio and return on the insurance portfolio should increase in the degree to which insurers face capital constraints. Figure 3 plots the average return on assets and return on insurance (ROI) in the U.S. combining all lines of business across all P&C insurers (left panel) and by RBC Capital ratio tercile (right panel). The ROI is computed as $100 * (1 - Combined\,Ratio)$, where the Combined Ratio is the ratio of claims plus other operational costs divided by premiums collected. Over the past 25 years, insurance operations generated a negative return of -1.2% on average, while the investment portfolio yielded 3.2% per year on average. The operational spread, defined as the difference between ROA and ROI, is higher for insurers that are more capital constrained.

[8] Changes in net worth affect total assets in the economy and therefore expected asset returns. Yet, an infinitesimal insurer takes expected asset returns as given, yielding a zero MPI in the unconstrained region.

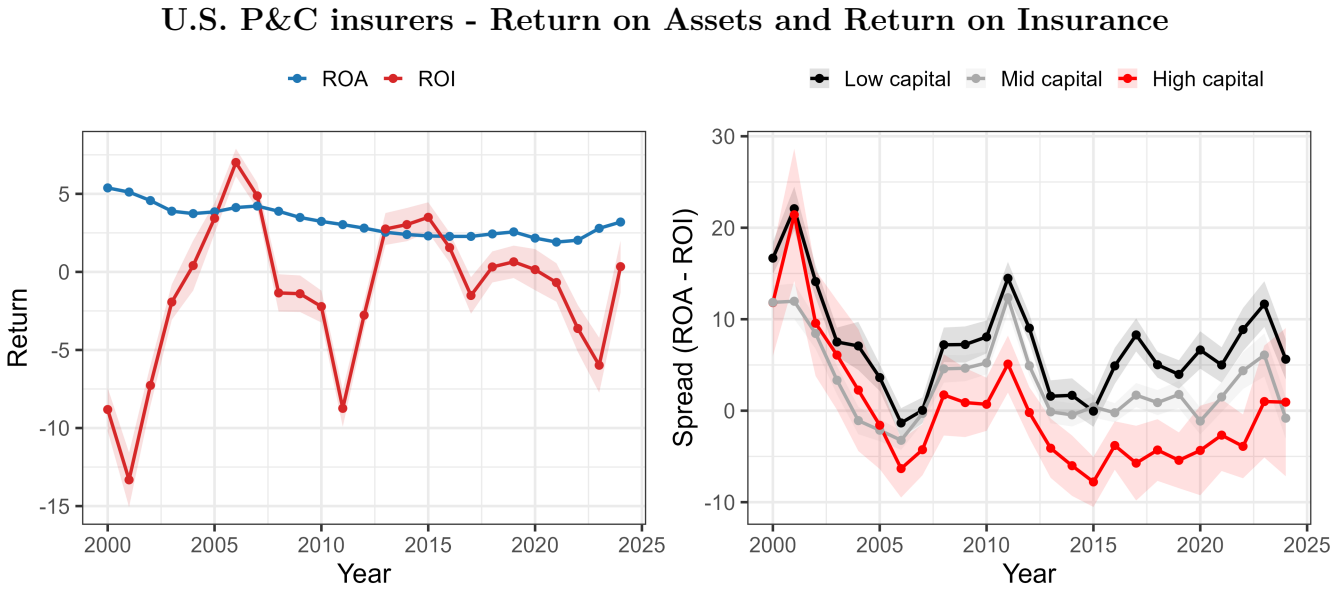



**Figure 3.** Return on Assets (ROA) and Return on Insurance (ROI) across all U.S. P&C insurers in all lines of business. The left panel shows the equal weight average of insurers' ROA and ROI by year. The right panel shows the spread, defined as the difference between ROA and ROI, by year and RBC ratio tercile.

In the constrained region, MPI is:

$$MPI^{con} = \frac{\partial \mathcal{X}_{t+1}(n,\theta)}{\partial n_t} = \frac{\mathbb{E}_t\left[\mathcal{M}(1+R^k_{t+1})\right]}{\zeta p^r - \mathbb{E}_t\left[\mathcal{M}\left(p^x_t(1+R^k_{t+1}) - \mathbf{1}_{t+1}\phi_{t+1}\right)\right] + \chi_1\chi_2 x^{\chi_2-1}_{t+1}} \tag{27}$$

The equation shows that MPI is increasing in expected asset returns and insurance prices, but decreasing in regulatory stringency and operational costs. High $\zeta$ reflects stringent capital regulation and indicates that for a given change in net worth, insurance supply contractions are small.

## 3.5 Final good firms

There is a continuum of measure one of perfectly competitive firms that produce the final good, $y_t$, and pay out wages to households, $w_t$. Final good firms use an identical constant returns to scale Cobb-Douglas production function with aggregate capital, $k_t$, and labor, $h_t$, as inputs:

$$y_t = z_t k_t^{\alpha} h_t^{1-\alpha}, \quad 0 < \alpha < 1 \tag{28}$$

where $z_t$ is aggregate productivity. Final good firms produce, pay wages to households and generate returns on assets for insurers. $z_t$ follows a Markov process. Capital of final good firms is not destroyed by disasters in the baseline model.

The capital-producing firm is cash-strapped and depends on insurers and external investors for external financing. Total assets in the economy equal:

$$\bar{a}_t = a_t^{out} + a_t^{ins} \tag{29}$$

where $a_t^{out}$ equals assets from outside investors and $a_t^{ins}$ equals assets coming from insurance companies.[9] Each capital-producing firm takes in claims $\bar{a}_t(j)$ and produces $\Phi(\bar{a}(j))$ new units of the capital good, where $\Phi(\cdot)$ is a transformation function such that $\Phi' > 0$ and $\Phi'' < 0$. The decision problem of each producer is thus:

$$\max_{\bar{a}_t(j)} \; q_t^{\bar{a}} \Phi\Big(\bar{a}_t(j)\Big) - \bar{a}_t(j) \tag{30}$$

Given symmetry across capital producers ($\bar{a}(j) = \bar{a}$), the aggregate price of capital is pinned down as follows:

$$q_t^{\bar{a}} = \Big[\Phi'\Big(\bar{a}_t\Big)\Big]^{-1} \tag{31}$$

Profits from capital goods producers are redistributed lump sum to households. Finally, the return on capital that insurers and outside investors take as given is given by:

$$R_{t+1}^k = \frac{z_{t+1}\alpha k_{t+1}^{\alpha-1} h_{t+1}^{1-\alpha}}{q_t^{\bar{a}}} \tag{32}$$

After being used in the production of the final good, capital fully depreciates.

# 4 Unregulated Decentralized Competitive Equilibrium

A recursive competitive equilibrium consists of the insurer value and policy functions, household policy functions, such that, given a vector of aggregate pricing $\{p^s, p^x, w, R^k\}$ functions and the perceived laws of motions for savings $\Gamma^B(B_t, X_t, n_t, \mathcal{S}_t)$, insurance $\Gamma^X(B_t, X_t, n_t, \mathcal{S}_t)$, and net worth $\Gamma^N(B_t, X_t, n_t, \mathcal{S}_t)$:

1. The value and policy functions of households and insurers solve their corresponding

[9] Considering that insurers offer insurance to cover houses of size $s = 1$, their size in the economy is modest. Introducing outside investors ensures that the fraction of equity held by insurers in the economy matches reasonable macro-economic averages.

decision problems.

2. The perceived law of motion for aggregate bonds and aggregate insurance, and perceived pricing functions are consistent with the actual decisions rules and actual prices, respectively.

3. The goods market clears: $c_t = y_t + b_t - b_{t+1}/R_t$[10], the insurance market clears, the housing market clears: $s_{t+1} = 1$ and the asset market clears: $k_{t+1} = \Phi(a_t)$

4. The household resource, collateral, insurer balance sheet and insurer capital regulation constraints hold.

# 5 Taking the Model to the Data

This section calibrates the model. I first motivate the choice of parameters and then discuss the model's performance in matching empirical moments.

## 5.1 Parameterization

Table 2 shows the parameter values used in the baseline model, which I motivate next.

**Disaster parameters** I calculate the disaster probability $p$ as the fraction of zip-codes that experienced positive catastrophic fire damages in the state of California during 2018-2023 (excluding indirect smoke damages from neighboring zip-codes), yielding an estimate of 0.0523. I compute the parameters $\rho_\theta$ and $\sigma_\theta$ based on county- and state-level disaster damages. Figure B.3 plots the distribution of auto-correlation coefficients in annual inflation-adjusted disaster damages at both state and county-level. The state- and county-level median autocorrelation are both near 0, although there exists a substantial number of geographies with either strongly negative or strongly positive autocorrelation. The median volatility of disaster damages is 0.9 at both the state and county level. These statistics suggest that the size of disaster damages is largely random and shows little persistence.

**Household parameters** I choose a simple equity redraw cost function $f(l_t) = \frac{\vartheta_l}{2} l^2$ and set $\vartheta_l$ so that average home equity redraws as a share of home values equal 14.1% (Freddie Mac, 2022). I set $\nu_s$ to match average housing costs as a share of total household spending

[10] Consumption thus equals total production net of foreign capital flows.

**Table 2**
Model Parametrization

| Parameter | Value | Description | Target/Source |
|---|---|---|---|
| | | *Households* | |
| $\psi$ | 2 | Risk aversion | Literature |
| $\beta$ | 0.97 | Discount factor | Literature |
| $\nu_s$ | 0.041 | Weight on housing utility | Housing to total exp. 0.33 |
| $\sigma_s$ | 0.9 | Housing utility concavity | - |
| $\{\kappa^H, \kappa^L\}$ | $\{0.95, 0.75\}$ | Collateral constraint states | Average LTV |
| $p^r$ | 0.81 | Housing rebuilding cost | Rebuilding cost / $p^s$ upon shock 0.625 |
| $\vartheta_l$ | 4.5 | Home equity redraw cost | $l$ / $p^s$ 0.14 |
| | | *Firms and Technology* | |
| $\alpha$ | 0.33 | Average Capital share | Literature |
| $\{z^L, z^H\}$ | $\{0.99, 1.01\}$ | Aggregate productivity states | Output volatility 0.01 |
| $a^{out}$ | 32 | Capital outside investors | 3.3% share P&C ins. in Corp. Bonds |
| $A_\Phi$ | 6.0 | Capital production, scale | Average investment yield 3.2% |
| $b_\Phi$ | 0.75 | Capital production, curvature | Jamilov and Monacelli (2025) |
| | | *Insurers* | |
| $\bar{\delta}$ | 0 | Target payout rate | Normalization |
| $\zeta$ | 0.95 | Capital constraints parameter | Share constrained insurers 27.6% |
| $\chi_1$ | 0.031 | Non-claim expenses, linear | Average MPI 0.5 |
| $\chi_2$ | 2 | Non-claim expenses, quadratic | Jamilov and Monacelli (2025) |
| | | *Stochastic Processes* | |
| $\pi^d$ | 0.0523 | Disaster probability | Estimated |
| $\rho_\theta$ | 0.0 | Persistence, disaster severity | Estimated |
| $\sigma_\theta$ | 0.9 | Volatility, disaster severity | Estimated |
| $\pi_z^{LL}$, $\pi_z^{HH}$ | 0.9, 0.9 | Productivity persistence | Literature |
| $\pi_\kappa^{HL}$, $\pi_\kappa^{LL}$ | 0.1, 0.0 | Collateral-regime transitions | Literature |
| $\rho_R$ | 0.68 | Persistence, Interest rate | Literature |
| $\sigma_\varsigma$ | 1.86% | Volatility, Interest rate | Literature |
| $\bar{R}$ | 2.0% | Mean, Interest rate | Literature |

of 0.33 (U.S. Bureau of Labor Statistics, 2025). Lastly, I set rebuilding cost $p^r$ to match the ratio of rebuilding costs to the market value of the house upon a disaster of 62.5% (Costa and Baker, 2024).[11]

**Insurer parameters** I set $\bar{a}^{out}$ to match the market share of P&C insurers in the corporate bond market of 3.3%, which is derived using the ownership data reported by Ge and Weisbach

[11] Costa and Baker (2024) estimate that following Californian wildfires replacement costs as a fraction of home values range between 45% and 80%.

(2021). I set $\zeta$ to match the fraction of insurers that are considered capital constrained. I empirically estimate the fraction of active insurers who have RBC ratios below 300, which is the point at which insurers start receiving regulatory scrutiny. I calibrate the dividends smoothing parameter to match the insurance gap of 45% in the U.S. (Aon, 2026).[12] I set $\chi_1$ to match empirical estimates of marginal propensity to insure. I estimate MPI through a simple OLS regression:

$$\Delta X_{izt} = \delta \Delta n_{it} + \eta_i + \eta_z \times \iota_t + \epsilon_{i,t}, \tag{33}$$

where $\Delta X_{izt} = \log X_{izt} - \log X_{izt-1}$ are insurer zip code level changes in sold exposures and $\Delta n_{it} = \log n_{it} - \log n_{it-1}$ captures changes in net worth. Zip-code-by-year fixed effects $\eta_z \times \iota_t$ capture local insurance shocks that affect insurers operating in this zip code. I therefore estimate how changes in net worth affect insurance quantities by comparing insurers within a zip-code year pair, i.e., by keeping local insurance conditions constant. Insurer fixed effects $\eta_i$ absorb time-invariant insurer traits, such as risk-taking, that could affect growth or shrinkage in the insurer's portfolio.

Equilibrium insurance quantities and net worth are jointly determined and an OLS estimate of $\beta$ would be subject to a simultaneity bias. However, zip-code level insurance revenues are small compared to overall insurer net worth. Hence, changes in zip-code level revenues are unlikely to individually materially affect insurer balance sheet conditions, which alleviates the simultaneity bias.

Panel A in Table A.2 shows that changes in insurer net worth are an important determinant of local insurance quantities. Empirical estimates of MPI range between 0.27 and 0.75. Panel B shows similar findings for insurer exit decisions: insurers with higher net worth levels and increases are less likely to exit zip-codes. I assume an average marginal propensity of 0.5 to match in the quantitative exercise.

Table A.3 investigates heterogeneity in MPI across insurers. Columns 1 and 2 show that insurers with low capital ratios and highly leveraged insurers have high MPI. This is consistent with insurers who are operating close to binding capital constraints experiencing larger volatility in their supply following changes in net worth. Column 3 indicates that insurers who have higher market shares tend to have lower MPI on average. The supply of insurers who enjoy local market power is thus less sensitive to net worth fluctuations.

[12] The insurance gap in the U.S. varies over time. Exhibit 16 in Aon (2026) shows that the average equals roughly 45%.

**Production parameters** Aggregate productivity $z_t$ follows a Markov chain with two states $\{z^L, z^H\}$ and corresponding transition probabilities $\{\pi_z^{LL}, \pi_z^{HH}\}$. The global interest rate $R$ follows an independent AR(1) process. The AR(1) process for the logged gross real interest rate is $\ln(R_t) = (1-\rho_R)\bar{R} + \rho_R \ln(R_{t-1}) + \varsigma_t$ with $\varsigma_t \sim \mathcal{N}(0, \sigma_\varsigma^2)$. The interest rate is discretized using Tauchen's quadrature method with three realizations.

Following Jamilov and Monacelli (2025), I assume that the elasticity of output with respect to capital is $\alpha = 0.33$ and that the capital production function takes the following form:

$$\Phi(\bar{a}) = A_\Phi \bar{a}^{b_\Phi}, \quad A_\Phi, b_\Phi > 0 \tag{34}$$

Parameter $b_\Phi$ is set to 0.75 so that the elasticity of the price of capital with respect to equity investments is equal to 0.25. I calibrate $A_\Phi$ so that the average return on capital $R^k$ in the baseline simulations is 3.2%.

## 5.2 Targeted and Untargeted moments

Table 3 presents the calibration results. The model matches most targeted moments, although the model predicts a lower average marginal propensity to insure and lower volatility of output.

I next evaluate how the model's simulation lines up against untargeted moments.[13] The model matches untargeted moments reasonably well, including housing damages as a fraction of GDP, the ROA-ROI spread on insurers' portfolios and the ratio of premium income to net worth. The model however predicts lower dividend payouts as a fraction of net worth.

# 6 Baseline Model Results

This section first presents the policy functions of the household and insurer. It then shows the Impulse Response Functions after a disaster shock under different levels of insurer net worth.

[13] I construct housing damages to GDP by dividing the average annual cost of disaster damages of $156.2 billion by the average GDP in the U.S. over the period 2021-2025 (Climate Central, 2026). I construct the ROA-ROI spread using the empirical estimates underlying Figure 3. I compute the premium to net worth ratio .

Finally, I compute dividend to net worth by dividing stockholder dividends by policyholders' surplus as reported in National Association of Insurance Commissioners (2025).

**Table 3**
Model Averages and Calibration Targets in the baseline economy

| | Model | Target / Data |
|---|---|---|
| *Targeted moments* | | |
| Insurance gap | 42.9% | 45% |
| Return on invested assets | 3.3% | 3.2% |
| Share insurers in corporate bonds | 2.8% | 3.3% |
| Marginal Propensity to insure | 0.24 | 0.50 |
| Share constrained insurers | 28.9% | 27.6% |
| Housing cost to total expenditures | 0.34 | 0.33 |
| Rebuilding cost to home value upon shock | 0.63 | 0.63 |
| Home equity redraw to home value | 0.14 | 0.14 |
| Output volatility $\sigma_y$ | 0.001 | 0.01 |
| *Untargeted moments* | | |
| Housing damages to GDP | 0.63% | 0.60% |
| ROA-ROI spread | 2.2% | 3.0% |
| XXXPremium income to net worth | 4.0% | 3.8% |
| Dividend payouts to net worth | 4.5% | 8.2% |

## 6.1 Policy functions

**Household policy functions** Figure 4 shows the policy function for the household as a function of the savings state $b$, separately for the no-disaster and disaster case. The disaster case is split by a low- and high insurance case to understand how debt and insurance states jointly shape decisions and outcomes upon disasters.

The top left panel shows that consumption is higher in the no-disaster case than in the disaster case. Upon a disaster, housing collateral gets destroyed and, when households are not fully insured, fewer resources are available for consumption. The drop in consumption is larger for the low-insurance case, highlighting the importance of insurance in preserving economic activity. Furthermore, the slope between consumption and savings is steeper in the disaster than in the no-disaster case. The non-linear relationship between consumption and savings in the disaster states reflects the housing fire sales dynamic. If households have high debt (negative savings), house prices drop more upon a disaster shock compared to when households have low debt. The steeper slope 'kicks in' earlier for low levels of insurance, highlighting the important interactions between disasters, debt and insurance in determining consumption.

Next-period's savings choice is an increasing linear function of current savings without

a disaster. In contrast, disasters destroy house prices through the fire-sale dynamic and erode borrowing capacity, especially in states of high debt (negative savings). Hence, for sufficiently low levels of savings, the slope between current savings and next-period savings turns negative. For low levels of savings today, next-period savings are higher for the low insurance case, given the lower borrowing capacity relative to the high insurance case.

Next-period insurance choice decreases in savings as higher savings lower the probability of experiencing binding collateral constraints in the future, reducing the need for insurance. Savings and insurance can therefore be viewed as substitutes within this framework. The change in the new insurance choice upon a disaster mimics the borrowing choice. A low-insurance state with a disaster destroys borrowing capacity and households have fewer resources to buy new insurance for the next period. Finally, optimal equity redraws are independent of savings without a disaster, but similarly drop upon a disaster.

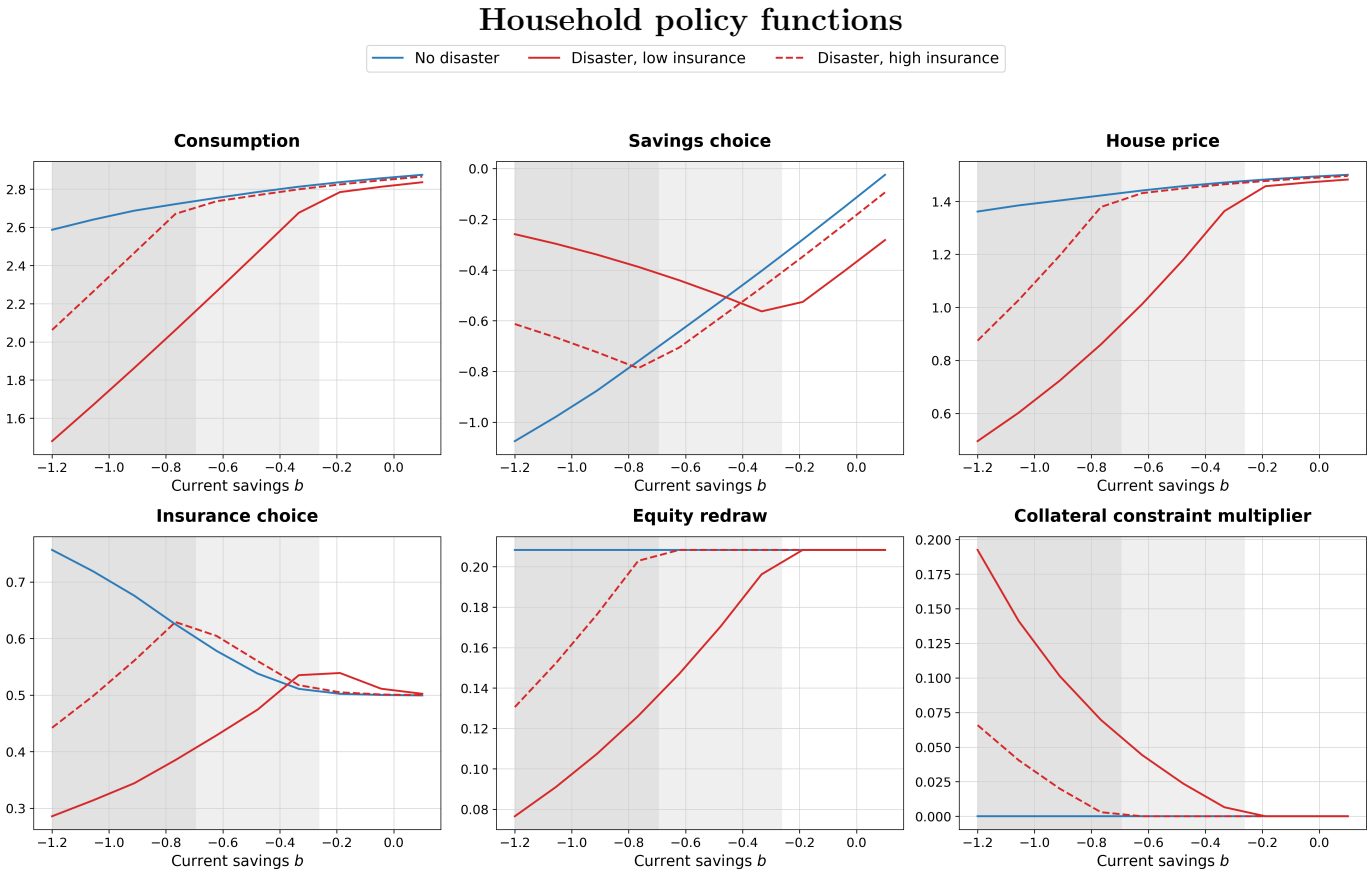



**Figure 4.** Household consumption, savings, house prices, insurance, equity redraws and the collateral constraint multiplier as function of outstanding savings in the decentralized competitive equilibrium. The blue line corresponds to the state where no disaster occurred. The red solid (dashed) line corresponds to a disaster state where the household insurance state is the highest (lowest) on the grid. $\kappa$ is set to its highest value. All other states are set to their median. The light (dark) shaded area represents the area in which the household with low (high) faces binding collateral constraints.

**Insurer policy functions** Figure 5 plots equilibrium insurance supply, insurance prices, marginal propensity to insure and equilibrium asset returns as a function of the insurer net worth state, and separately for the disaster- and no-disaster state. The shaded area reflects the area in which insurers are experiencing binding capital constraints. Insurance and savings states are set to the median, and the collateral constraint parameter $\kappa$ is set to its highest value.

There are several observations to make. First, in the constrained region, insurance supply sharply rises in net worth, whereas it is independent of net worth in the unconstrained region. When capital constraints bind, insurers are dependent on net worth to grow their insurance portfolio and MPI is positive. When unconstrained, insurers optimize over insurance supply independently of net worth. The only variation of insurance supply across net worth in the unconstrained region reflects endogenously changing equilibrium insurance prices and expected asset returns, with asset returns decreasing slightly in the amount of assets in the economy.

Second, MPI is high when insurers are constrained and zero otherwise. When unconstrained, insurers' net worth changes do not affect equilibrium insurance quantities. When constrained, insurance supply drops sharply upon shocks. Such supply reductions can be costly for homeowners as the ability to insure against future disasters lowers and insurance prices increasingly reflect financial conditions of the insurer. Section 7 discusses how a social planner can appropriately tax or subsidize insurance to limit the frequency of visiting these states.

Third, equilibrium insurance prices are a decreasing function of insurer net worth. Insurance prices are high when insurers have low net worth and are constrained, consistent with the empirical evidence presented in Table A.4. The insurance prices shown in Figure 5 are conditional on debt and insurance states. As the simulations will display in section 6.2, insurance prices tend to drop upon disasters as disasters reduce debt. Lower leverage in turn reduces the willingness to pay for insurance as savings and insurance work as substitutes.

**Transmission of insurance market conditions into house prices** Figure 6 highlights how insurance market conditions affect welfare and house prices. The left panel shows that household welfare is increasing in net worth, consistently across different levels of savings. Insurer net worth is beneficial for households as it keeps insurance affordable and through more insurance take-up prevents future sharp reductions in consumption. The right panel shows that house prices increase in insurer net worth, and the shape mimics that of the relationship between insurance prices and net worth (see Figure 5). House prices increase

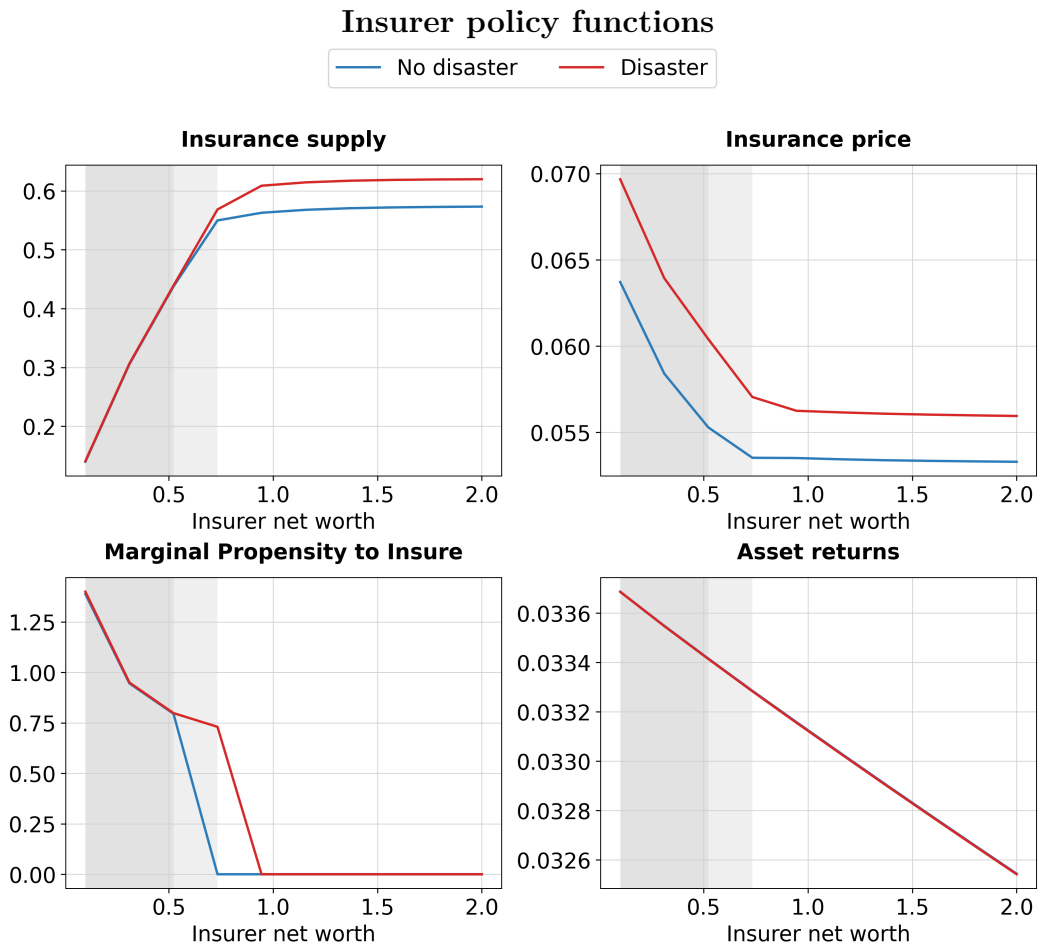



**Figure 5.** Insurer assets, insurance supply and insurance prices as a function of insurer net worth. The blue line corresponds to the state where no disaster occurred. The red solid (dashed) line corresponds to a disaster state where the household insurance state is the highest (lowest) on the grid. Shaded areas reflect net worth states where capital regulation binds. Insurance and savings states are set to the median, and collateral constraints parameter $\kappa$ is set to its highest value.

in insurer net worth through the direct effect of insurance premiums on marginal utility, affecting house valuations, but also through the expectation of future house prices that depends on future insurance coverage.

I next exploit the variation of insurance and house prices across the net worth grid to estimate the elasticity of house prices with respect to insurance prices. Keeping other states fixed, I obtain a mean elasticity of -0.13 and a median elasticity of -0.10, which aligns with the empirical estimates in Table 1.

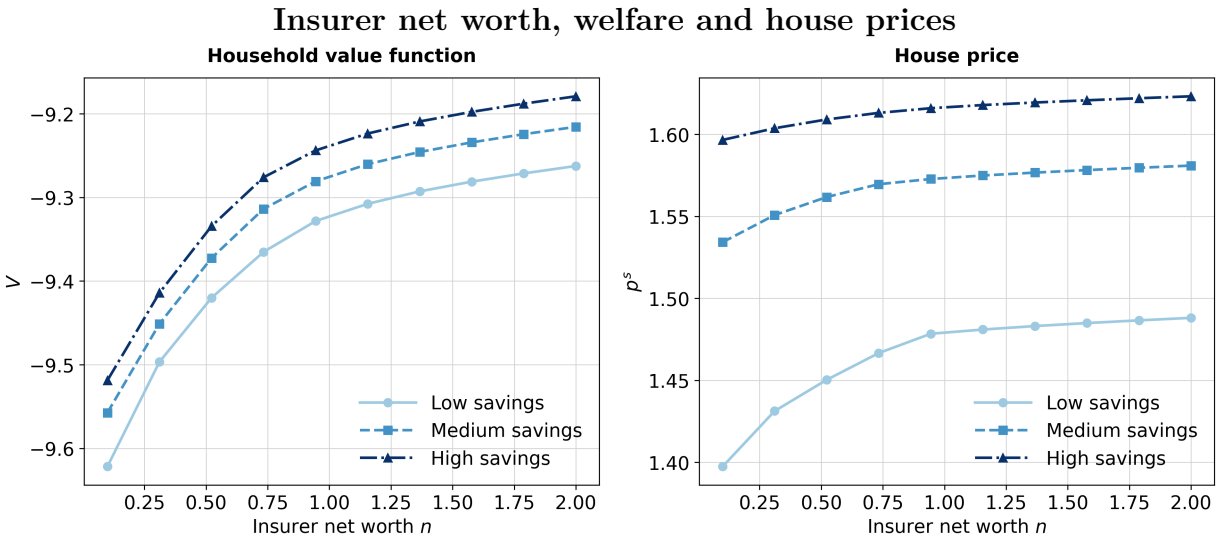



**Figure 6.** The household value function in the decentralized equilibrium and house prices as a function of insurer net worth for different states of savings.

## 6.2 Impulse Response Functions

I study impulse response functions (IRF) to a disaster shock under different levels of insurer net worth and different values of the collateral constraint parameter $\kappa$. Different levels of insurers' net worth imply different equilibrium insurance quantities and consequently differing fragilities of the housing market to disaster shocks. Different values of $\kappa$ imply that the collateral constraint binds to different degrees, which leads to different deleveraging needs and house price dynamics upon a disaster shock. This setup allows me to assess whether the model can replicate the joint role of insurance and LTV ratios in driving house price reductions upon disasters as empirically identified in Figure 2.

To obtain these IRFs, I first run a simulation for T=2000 periods and discard the first 100 periods. At time $T^* < T$ a random disaster shock occurs. I run a second simulation that is identical to the first except that there is no disaster shock. The impulse response is constructed by subtracting the second from the first simulation. In the fragile economy simulation, a negative financial shock to insurers occurs at $T^* - 1$, which reduces their net worth and therefore equilibrium insurance quantities.[14] The fragile economy reflects the possible scenario in which insurers are recurrently hit by disasters or broader economic shocks, shrinking their risk-bearing capacity, or when uncertainty is high so that effective risk-taking by insurers decreases. Similarly, I construct the IRF for the fragile-insurer economy by subtracting its simulation from the counterfactual simulation with the same negative shock to

[14] Specifically, I reduce the net worth by 20% of the range of net worth grid.

net worth but without the disaster. I repeat this procedure for the case where the disaster shock coincides with a shock to the collateral constraint parameter $\kappa$.

Figure 7 presents the results. A disaster shock significantly reduces consumption and house prices in the baseline economy, by 8% and 15% respectively, when $\kappa$ is high (black solid line). Fragile insurers amplify this response, with consumption and house prices dropping by 12% and 20%, respectively (orange solid line). The state of the insurance market is an important determinant of house price fluctuations upon disasters. When disaster shocks coincide with tightening collateral constraints, shocks to consumption and house prices are largest, amounting to 20% and 32% respectively (orange dashed line). The model can thus match the non-linear interaction of insurance coverage and collateral constraint tightness in house price reductions upon shocks. The effects on consumption and house price drops are smaller than in Bianchi and Mendoza (2018). Part of this can be explained by the fact that aggregate production capacity in this paper's economy remains largely unaffected as the capital provision of outside investors is not affected by disasters, whereas output in Bianchi and Mendoza (2018) drops next to asset prices.

Savings increase sharply by 10-40% upon disasters depending on the state. Disaster shocks erode borrowing capacity upon a disaster shock, and households' ability to smooth consumption weakens, and this effect is larger in states with low insurance coverage and tight collateral constraints.

Equilibrium insurance quantities drop upon the shock and are accompanied by lower insurance prices. While conditional on the debt, insurance and exogenous shock state, insurance prices decrease in net worth, as displayed in Figure 5, insurance prices decrease in savings because in high-savings states the probability of future collateral constraints becoming binding lowers, reducing the need for insurance take-up, as displayed in Figure 4. The household valuation channel dominates the insurer net worth channel so that insurance prices drop upon impact. Finally, as insurer net worth lowers upon impact, marginal propensity to insure increases. As insurers' net worth decreases, they are more likely to operate under binding capital constraints, which increases their sensitivity of insurance supply to net worth changes upon future shocks. When insurance price decreases are severe, MPI may also temporarily decrease.

To better understand the role of endogenous price adjustments and insurer deleveraging in driving insurance quantities after a disaster shock, I plot the reduction in insurance coverage after the shock depending on the insurer's net worth prior to the shock in Figure 8. I distinguish between insurance prices adjusting endogenously (dashed orange line) and

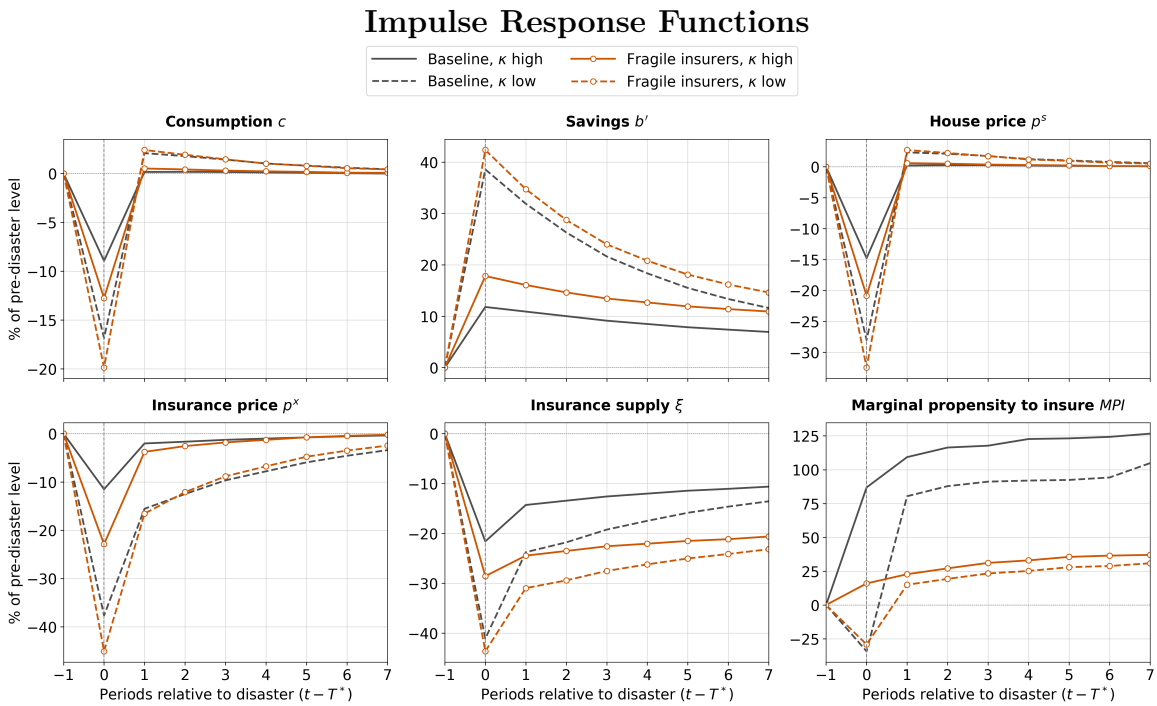



**Figure 7.** Impulse Response Functions to a Disaster Shock at $T^*$ in the baseline and a fragile economy under different values of the collateral constraint tightness parameter $\kappa$. In the fragile economy, insurer net worth is exogenously lowered prior to the arrival of the disaster shock at time $T^* - 1$. The black and orange lines correspond to the baseline and fragile economy, respectively. The dashed lines show the responses when the disaster shock coincides with a shock to collateral constraint tightness.

insurance prices remaining fixed to the pre-shock level (black solid line). Supply reductions become significantly steeper when insurer net worth prior to the disaster is sufficiently low so that upon a disaster the insurer becomes capital constrained. The shaded grey area depicts this threshold level of net worth.

Even when the insurer does not become constrained upon a disaster shock, insurance coverage may fall as insurance prices fall endogenously. When prices are assumed to be fixed, insurance coverage does not change, consistent with zero marginal propensity to insure in the unconstrained region (black line). Yet, as insurance prices drop upon impact, insurance coverage lowers, even when capital constraints are not binding. The model predicts that when insurers have significantly low net worth prior to the disaster, insurance prices may increase upon the shock.

# 7 Time-consistent insurance take-up

The results of the decentralized economy show that insurance is important for preserving collateral values and consumption upon disasters. Yet, high insurance coverage constrains

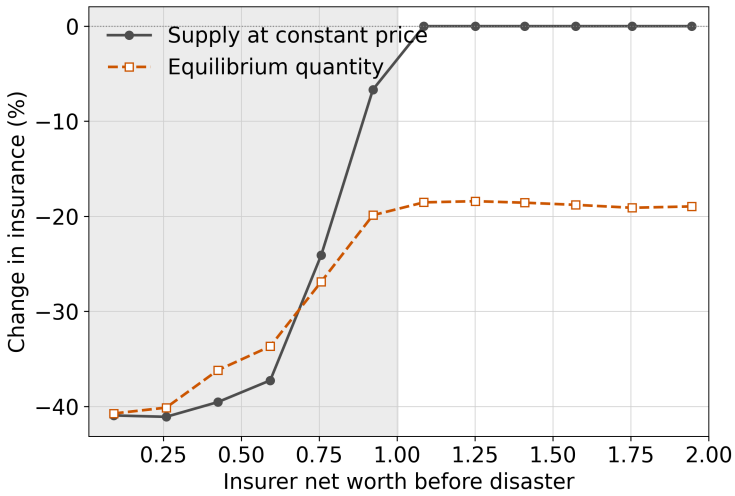



**Figure 8.** Percentage deviation in equilibrium insurance take-up depending on initial insurer net worth. The orange line plots the IRF when insurance prices adjust endogenously, while the black line assumes insurance prices are equal to the price level prior to the disaster shock occurring. The shaded area reflects the level of net worth prior to the disaster at which insurers become capital constrained following a disaster.

intermediaries by making them more operationally leveraged, which steepens insurance contractions upon a disaster. This section introduces a social planner who optimizes this intertemporal tradeoff.

In the DE, households take house prices as given. Consequently, they do not internalize how insurance decisions today affect house prices upon a disaster shock when the collateral constraint becomes binding. With low insurance coverage, disasters drive sharp reductions in house prices and consequently borrowing capacity and consumption. Similarly, households take insurer net worth as given and do not internalize how high insurance demand today affects future insurance prices and availability after disaster shocks when insurers' capital constraints become binding. The decentralized equilibrium therefore is characterized by two inefficiencies that a planner can internalize to potentially improve long-run welfare. Ex ante, it is not clear whether the planner seeks to tax or subsidize insurance as it depends on the strength of these channels.

Section 7.1 presents the planner's problem and characterizes the externalities. Section 7.2 outlines how the planner's allocation can be decentralized using a Pigouvian tax on borrowing and insurance. Section 7.3 quantifies the social planner's problem and compares the planner's solution to the decentralized equilibrium quantitatively.

## 7.1 Planner's problem

I formulate the social planner's problem similar to the primal approach to optimal policy analysis. The planner chooses insurance $x_{t+1}$ and $b_{t+1}$ on behalf of the household subject to feasibility constraints, including the budget, collateral and market clearing constraints, and an implementability constraint, which concerns the housing Euler equation (13). Since the planner cannot set prices directly, the planner is required to choose an allocation that satisfies the housing Euler equation. The Euler and insurance optimality equations of the DE do not enter the planner's problem as constraints because savings and insurance are controlled by the planner and their valuation may be different than the household's private valuation.

When the planner can influence prices directly, the insurance market-clearing condition enter the planner's problem as a feasibility constraint because every combination of prices and insurance take-up should match what the insurer would supply at this price. The Lagrange parameter on the market clearing constraint reflects the price impact of demand on insurance prices and for how higher insurance prices recapitalize the insurer, evaluated at the social value of insurer net worth. When the planner is assumed to be a price-taker, market-clearing is satisfied without this constraint and therefore does not enter the problem. I develop and solve for equilibria with both a price taking planner and a planner who can influence prices.

Households no longer decide on $b_{t+1}$ and $x_{t+1}$ and the household therefore takes as given the borrowing transfer $T_t^b = b_t - \frac{b_{t+1}}{R_t}$ and the net insurance transfer $T_t^x = \mathbf{1}_t \phi_t x_t p^r - p_t^x x_{t+1}$. As housing is in fixed supply, the social planner does not optimize over the housing choice. Households continue to choose home equity redraws optimally. While it would enter the planner's problem as a constraint, it never binds and can therefore be omitted.

The planner lacks the ability to commit to future policies. Hence, following Bianchi and Mendoza (2018) and Biljanovska et al. (2026), I focus on obtaining Markov stationary policy rules expressed as a function of the state variables $(b, x, n, \mathcal{S})$. The social planner takes as given the future planners' choices. A time-consistent equilibrium is characterized by a fixed point at which the current decisions of the social planner coincide with the planner's future decisions.

Denote $\mathcal{X}(b, x, n, \mathcal{S})$ and $\mathcal{B}(b, x, n, \mathcal{S})$ as the policy rules in the constrained-efficient equilibrium, and $\mathcal{C}(b, x, n, \mathcal{S})$, $l(b, x, n, \mathcal{S})$, $\mathcal{P}^s(b, x, n, \mathcal{S})$, $\mathcal{P}^x(b, x, n, \mathcal{S})$, $\mu^*(b, x, n, \mathcal{S})$ and $\Psi^*(b, x, n, \mathcal{S})$ the associated functions that return the values under that policy rule. The social planner's problem can be represented in recursive form:

$$
\begin{aligned}
\mathcal{V}(b,x,n,\mathcal{S}) &= \max_{c,l,x',b',p^s,p^x} u(c,1) + \beta\,\mathbb{E}_{\mathcal{S}'|\mathcal{S}}\mathcal{V}(b',x',n',\mathcal{S}') \\
c + \frac{b'}{R} + p^x x' &= b + w + \mathbf{1}\phi\left[xp^r - 1\right] + l - f(l) \\
-\frac{b'}{R} &\leq \kappa\left(p^s\left(1-\mathbf{1}\phi\right) + \mathbf{1}\phi x p^r - l\right) \\
\mathcal{X}(p^x,n,\mathcal{S}) &= x' \\
p^s u'(c,1) &= \beta\,\mathbb{E}\Bigg[u'\Big(c'(b',x',n',\mathcal{S}'),\,1\Big) \\
&\qquad \times \left\{\nu_s g'(1) + p^{s'}(b',x',n',\mathcal{S}')\right\} \\
&\quad + \left[p^{s'}(b',x',n',\mathcal{S}')\left(1-\mathbf{1}'\phi'\right)\right] \times \mu'(b',x',n',\mathcal{S}')\kappa'\Bigg]
\end{aligned}
\tag{35}
$$

The planner economy's budget constraint has the multiplier $\lambda^* \geq 0$. The collateral constraint now has multiplier $\mu^* \geq 0$, which may be different from the private value of relaxing the constraint $\mu$. The multipliers on the market-clearing and implementability constraint are given by $\Psi^* \geq 0$, and $\xi^* \geq 0$, respectively. While the planner takes as given the decisions of the future planners, it internalizes how current choices affect the planner's future decisions in the right hand side of implementability and feasibility constraints, resulting in time consistent policy rules.

The planner acknowledges that the debt and insurance choice directly affect asset prices through current marginal utility. In addition, they affect asset prices indirectly by affecting the future planner's bond holdings and insurance coverage and insurer net worth, with their associated future allocations and prices.

The solution to the planner's problem is defined by the policy functions $b'(b,x,n,\mathcal{S})$, $x'(b,x,n,\mathcal{S})$ and their associated decision rules $c(b,x,n,\mathcal{S})$, $l(b,x,n,\mathcal{S})$, $\mu^*(b,x,n,\mathcal{S})$, $\Psi^*(b,x,n,\mathcal{S})$, pricing functions $p^s(b,x,n,\mathcal{S})$ and $p^x(b,x,n,\mathcal{S})$, and value function $V(b,x,n,\mathcal{S})$, the decision rules of future planners $\mathcal{B}(b,x,n,\mathcal{S})$ and $\mathcal{X}(b,x,n,\mathcal{S})$, and their associated decisions rules $\mathcal{C}(b,x,n,\mathcal{S})$, $\mathbf{l}(b,x,n,\mathcal{S})$, $\mu^*(b,x,n,\mathcal{S})$, $\mathbf{\Psi}^*(b,x,n,\mathcal{S})$, and asset prices $\mathcal{P}^s(b,x,n,\mathcal{S})$ and $\mathcal{P}^x(b,x,n,\mathcal{S})$, such that the following conditions hold:

- Planner's optimization: the value function $V(b,x,n,\mathcal{S})$, and the decisions rules $b'(b,x,n,\mathcal{S})$, $x'(b,x,n,\mathcal{S})$, $c(b,x,n,\mathcal{S})$, $l(b,x,n,\mathcal{S})$, $\mu^*(b,x,n,\mathcal{S})$, $\Psi^*(b,x,n,\mathcal{S})$, $p^s(b,x,n,\mathcal{S})$, $p^x(b,x,n,\mathcal{S})$ solve the dynamic problem in (35) given the planner's future allocations $\mathcal{B}(b,x,n,\mathcal{S})$, $\mathcal{C}(b,x,n,\mathcal{S})$, $\mathbf{l}(b,x,n,\mathcal{S})$, $\mu^*(b,x,n,\mathcal{S})$, $\mathbf{\Psi}^*(b,x,n,\mathcal{S})$, $\mathcal{P}^s(b,x,n,\mathcal{S})$, $\mathcal{P}^x(b,x,n,\mathcal{S})$.

- Time consistency: The current planner's optimal policy coincides with the conjectured policy rules governing the decisions of future planners, $b'(b,x,n,\mathcal{S}) = \mathcal{B}(b,x,n,\mathcal{S})$, $c(b,x,n,\mathcal{S}) = \mathcal{C}(b,x,n,\mathcal{S})$, $l(b,x,n,\mathcal{S}) = \mathbf{l}(b,x,n,\mathcal{S})$, $\mu^*(b,x,n,\mathcal{S}) = \mu^*(b,x,n,\mathcal{S})$, $\Psi^*(b,x,n,\mathcal{S}) = \mathbf{\Psi}^*(b,x,n,\mathcal{S})$, $p^s(b,x,n,\mathcal{S}) = \mathcal{P}^s(b,x,n,\mathcal{S})$, and $p^x(b,x,n,\mathcal{S}) = \mathcal{P}^x(b,x,n,\mathcal{S})$ for all $(b,x,n,\mathcal{S})$.

### 7.1.1 Comparison of equilibria

Taking the first-order conditions yields the following social planner's optimality conditions:

$$c_t :: \qquad u'(t) = \lambda_t^* + u''(t)\xi_t^* p_t^s \tag{36}$$

$$b_{t+1} :: \qquad \lambda_t^* = \beta R_t \mathbb{E}_{\mathcal{S}_{t+1}|\mathcal{S}_t}\left(\lambda_{t+1}^* + \xi_t^* \Omega_{t+1}^{b_{t+1}}\right) + \mu_t^* \tag{37}$$

$$x_{t+1} :: \qquad p^x \lambda_t^* + \Psi_t^* = \beta \mathbb{E}_{\mathcal{S}_{t+1}|\mathcal{S}_t}\left[\lambda_{t+1}^* + \mu_{t+1}^* \kappa_{t+1})\mathbf{1}_{t+1}\,\phi_{t+1} + V_{N_{t+1}} \Lambda_{x_{t+1}}^{N_{t+1}} + \xi_t^* \Omega_{t+1}^{x_{t+1}}\right] \tag{38}$$

$$p^x :: \qquad \lambda_t^* x_{t+1} = \Psi_t^* \mathcal{X}_t^{p^x}(p^x, N, \mathcal{S}) + \beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}}\left[\xi_t^* \Omega_{t+1}^{p^x} + V_{t+1}^{N_{t+1}}\, \Lambda_{p_t^x}^{N_{t+1}}\right] \tag{39}$$

$$p^s :: \qquad \xi_t^* = \frac{\kappa_t \mu_t^* (1 - \mathbf{1}_t \phi_t)}{u'(t)} \tag{40}$$

where

$$\begin{aligned} \Lambda_{x'}^{n'} &= p^x(1 + R'^k) - \mathbf{1}'\phi' p^r - \chi_1 \chi_2 x'^{\chi_2 - 1} \\ \Lambda_{p^x}^{n'} &= x_{t+1}(1 + R'^k) \end{aligned} \tag{41}$$

is the change in next-period insurer net worth given a small change in insurance take-up $x_{t+1}$ and insurance prices $p^x$, respectively.

Compared to the first-order for consumption in the DE, the SP condition (36) has an additional term $u''(t)\xi_t^* p_t^s < 0$, which captures how an increase in consumption today affects house prices through the righthandside of the housing implementability constraint. An increase in consumption directly raises asset prices and lowers the shadow cost of consumption.

Equation (37) has two additional terms worth mentioning. First, next to $\xi_t^* u''(t) p_t^s$ that is embedded in $\lambda_t^*$, the term $\lambda_{t+1}^* = u'(t+1) - \xi_{t+1} u''(t+1) p_{t+1}^s$ inside the expectation captures how borrowing today lowers consumption tomorrow and indirectly lowers house prices by increasing the marginal utility of consumption at $t+1$. The planner thus considers how high leverage affects house prices in all possible future states. As the additional term $\xi_{t+1}^* u''(t+1) p_{t+1}^s$ is negative, the marginal benefit of savings increases, and the planner's debt choice lowers. Second, the term $\Omega_{t+1}^{b_{t+1}}$ captures how borrowing today affects the future

planner's choice through the right-hand side of the implementability constraint and thus is a term to guarantee time-consistent decision making.[15]

Equation (38) is the planner's optimality condition for insurance and has a similar structure as the insurance optimality condition in the DE but differs in important ways. First, the term $-p^x \xi_t^* u''(t) p_t^s$ on the lefthandside (through $\lambda_t^*$) captures how insurance costs change marginal utility of consumption today and indirectly affect house prices.

Second, similar to the debt choice, the term $\xi_{t+1} u''(t+1) p_{t+1}^s \mathbf{1}_{t+1} \phi_{t+1}$ (through $\lambda_{t+1}^*$) captures how insurance today preserves consumption in the event of a disaster and indirectly affects tomorrow's house prices through tomorrow's marginal utility and collateral constraints. As in the DE, the marginal benefit of insurance is proportional to the value of relaxing borrowing constraints in the future $\mu_{t+1}^*$. Quantitatively, I find that the social value $\mu_{t+1}^*$ exceeds the private value $\mu_{t+1}$.

Third, the marginal cost of insurance increases with $\Psi_t^*$. While the household in the DE is atomistic and does not consider how individual insurance demand affects equilibrium prices, the planner chooses aggregate insurance take-up. The planner recognizes that high insurance take-up makes the insurer constrained, through either capital market frictions or convex operational costs, which makes the equilibrium insurance price reflect the insurer's financial constraints. When $\Psi^*$ is high, the marginal cost of insurance increases, and equilibrium insurance prices lower.

Fourth, $V_{N_{t+1}} \Lambda_{x_{t+1}}^{N_{t+1}}$ captures the marginal value of insurers' future net worth with respect to the insurance choice. When the planner values accumulating insurer net worth, the planner may choose to increase the insurance price to recapitalize the insurer. Simultaneously, high insurance prices lead to high equilibrium insurance quantities that increase the operating leverage of the insurer and therefore insurance market crash-risk. Whether the planner's marginal value of net worth is positive or negative depends on the strength of these two channels.

Fifth, $\Omega_{t+1}^{x_{t+1}}$ captures how insurance take-up affects the future's planner choices through the righthandside of the implementability constraint. For its policy to be time-consistent, the planner considers how its choice affects future planner dynamically.[16]

[15] As shown in the Appendix, this term consists of how borrowing today affects: i) future marginal utility and thus discounting of future house prices, 2) equilibrium house prices upon shocks through possible fire sales, and 3) collateral constraint tightness at time $t+1$.

[16] The appendix decomposes $\Omega_{t+1}^{x_{t+1}}$ into four terms. The first term reflects how insurance affects future marginal utility and therefore future asset prices. Since higher $x'$ raises $c'$ in the event of a disaster, and given $u''(c) < 0$, this term is positive. The second term captures how insurance holdings affect asset prices and insurance payouts. Insurance take-up increases collateral values in the event of a disaster and reduces fire sales. The third term captures how insurance holdings affect the tightness of the collateral constraint at

Overall, the planner's marginal benefit of insurance depends on how insurance protects wealth in the event of a disaster and how insurance dynamically affects the net worth of the insurers evaluated at its social value.

Equation (39) shows the first-order condition with respect to the insurance price. The left hand side captures the marginal cost of increasing the insurance price, valued at marginal utility $\lambda^*$ multiplied by the insurance quantity. Increasing insurance prices is costly as it lowers consumption proportional to the insurance coverage. The right hand side captures the marginal benefit of increasing insurance prices, which is given by the slope of supply with respect to the insurance price multiplied by the Lagrange parameter $\Psi_t^*$. When households pay more for insurance, the equilibrium insurance quantity increases, and the benefit of this is given by $\Psi^*$. $\Psi_t^*$ reflects differences in the valuation of insurance between the household and the insurer. When $\Psi_t > 0$, the planner perceives the insurance price as too costly relative to the benefit of capitalizing insurers. In this case, the planner seeks to lower insurance prices until the marginal cost of raising insurance prices equals the marginal benefit of funding the insurer. When capital constraints bind, insurers' supply is inelastic and the planner may be unable to set $\Psi_t = 0$. The Appendix shows that equations (38) and (39) allow for identifying $p^x$ and $\Psi^*$ jointly.

Finally, equation (40) captures the first-order conditions with respect to house prices. As in Bianchi and Mendoza (2018), the value of relaxing the implementability constraint is proportional to the value of relaxing the collateral constraint. Equations (36) and (40) jointly show that the planner's consumption affects house prices and consequently how severely the implementability constraint binds.

## 7.2 Pigouvian tax on insurance

Appendix 10.4 shows that the social planner's allocation can be decentralized with a tax or subsidy on insurance, which equals the ratio of the insurance price in the DE to that in the planner's economy minus 1:

$$\tau^x = \frac{(p_t^x)^{DE}}{(p_t^x)^{PL}} - 1 \tag{42}$$

When the planner values insurance more than in the decentralized economy, leading to relatively high insurance prices, the tax becomes negative. Note that both the numerator

time $t+1$, which in turn affects house prices at time $t$. This term is negative. Fourthly, the planner considers how insurance take-up affects house price through the future net worth of the insurer. Quantitatively, the model shows that house prices are higher when insurers have high net worth, driven by contemporaneous insurance prices and future expectations in these states.

and denominator are evaluated at the planner's policy functions as when the tax is implemented the planner's and decentralized equilibrium's insurance choice coincide. Whenever the implementability and feasibility constraints today do not bind, one can derive the macro-prudential (MP) insurance tax:

$$\tau^x_{MP} = \frac{\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ (u'(c',1) + \mu' \kappa') \mathbf{1}' \phi' p^r \right]}{\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ (\lambda'^* + \mu'^* \kappa') \mathbf{1}' \, \phi' p^r + V_{N'} \, \Lambda^{N'}_{x'} \right]} - 1 \tag{43}$$

The macro-prudential (MP) insurance tax is levied in good times when household collateral constraints do not bind, but when constraints may bind in the future with positive probability. The macro-prudential tax consists of two parts. First, the tax depends on how the planner and the household in the DE differently value the benefit of insurance upon a disaster. Both the numerator and the denominator share the term $\mathbf{1}'\phi' p^r$, which captures the insurance payout at a disaster with full insurance. Importantly, the household in the DE and planner value future claim differently. Whenever $\lambda'^* + \mu'^* \kappa' > u'(c',1) + \mu' \kappa'$, the marginal benefit of insurance in the planner economy exceeds that of the household in the DE, calling for a subsidy.

Second, the macro-prudential tax depends on how insurance take-up today marginally affects insurers' future net worth, captured by $\Lambda^{N'}_{x'}$ evaluated at its social value $V_{N'}$. Note that $\Lambda^{N'}_{x'}$ is evaluated across all possible future states. In disaster states $\Lambda^{N'}_{x'}$ is negative as more insurance leads to insurers paying out more claims, while $\Lambda^{N'}_{x'}$ is positive in no disaster states as more insurance funds the asset base that generates a non-negative return $R'^k$. If losing net worth in disaster states is significantly more costly than gaining net worth in no disaster states, $\mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ V_{N'} \, \Lambda^{N'}_{x'} \right]$ may be negative, calling for an insurance tax.

The macro-prudential insurance tax thus broadly consists of two components. One component internalizes how insurance affects house prices upon a disaster shock by mitigating fire sales. The second component internalizes how more insurance today could either recapitalize the insurer or lead to possible binding insurer capital constraints in the future.

Similarly, one can derive the optimal tax on borrowing, which is similar to the tax derived in Bianchi and Mendoza (2018) and Biljanovska et al. (2026):

$$1 + \tau^b = \frac{\beta R \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left( u'(c',1) - \xi'^* u''(c',1) p'^s + \xi^* \hat{\Omega}'_{b'} \right) + \xi^* u''(c,1) p^s}{\beta R \mathbb{E}_{\mathcal{S}'|\mathcal{S}} u'(c',1)} \tag{44}$$

## 7.3 Quantitative analysis of the social planner

**Policy functions** Figure 9 compares the policy functions of the planner with those in the decentralized equilibrium. I distinguish between equilibria in which the planner is a price-taker ($\Psi^* = 0$) or a price-setter ($\Psi^* \neq 0$).

The insurance choice absent a disaster of the price-taking planner is higher than that of the household in the DE, and the difference becomes larger for low levels of savings (high levels of debt). When more indebted, the possibility of experiencing future binding collateral constraints is high. The value of insurance increases in these states as the planner internalizes possible house price crashes that lead to severe consumption drops.

In contrast, the insurance choice of the price-setting planner is lower than in the DE. When the insurer's supply curve is sufficiently steep, small changes in insurance quantities translate into large changes in insurance prices. When the reduction in marginal cost of insurance exceeds the reduction in marginal benefit of future net worth, the price-setting planner has an incentive to continue reducing prices and de-fund the insurer, which translates into a lower equilibrium insurance take-up. When prices lower in a given state, the attractiveness for insurers to operate in the market lowers and available net worth lowers, amplifying the financial constraints in the insurance market. As the financial frictions component in the insurance price grows, the incentive for the planner to lower insurance prices grows further, so that financial constraints and low insurance demand become self-reinforcing.

The new savings choice of both planners in the no disaster state is higher than in the DE. The planner chooses lower debt as it internalizes how high leverage negatively affects house prices upon disasters. While differences are small, the non-linear interaction between debt and house prices mean that these changes can have a meaningful effect on how borrowing capacity and consumption changes upon a disaster.

**Taxation** Figure 10 plots the macro-prudential insurance and debt tax for the price-taking planner across the savings state, and separately for different levels of net worth. Consistent with Bianchi and Mendoza (2018), the macro-prudential debt tax is exclusively positive and decreases in savings. The level of debt taxes is lower than in Bianchi and Mendoza (2018), who find an optimal debt tax of up to 8%. As disaster risk can be mitigated by both lower debt and more insurance, when both corrections are at play, the magnitude of only the debt correction is smaller.

Figure 11 decomposes the macro-prudential insurance tax into the parts stemming from differences in how the household in the DE and planner value future insurance claims (blue

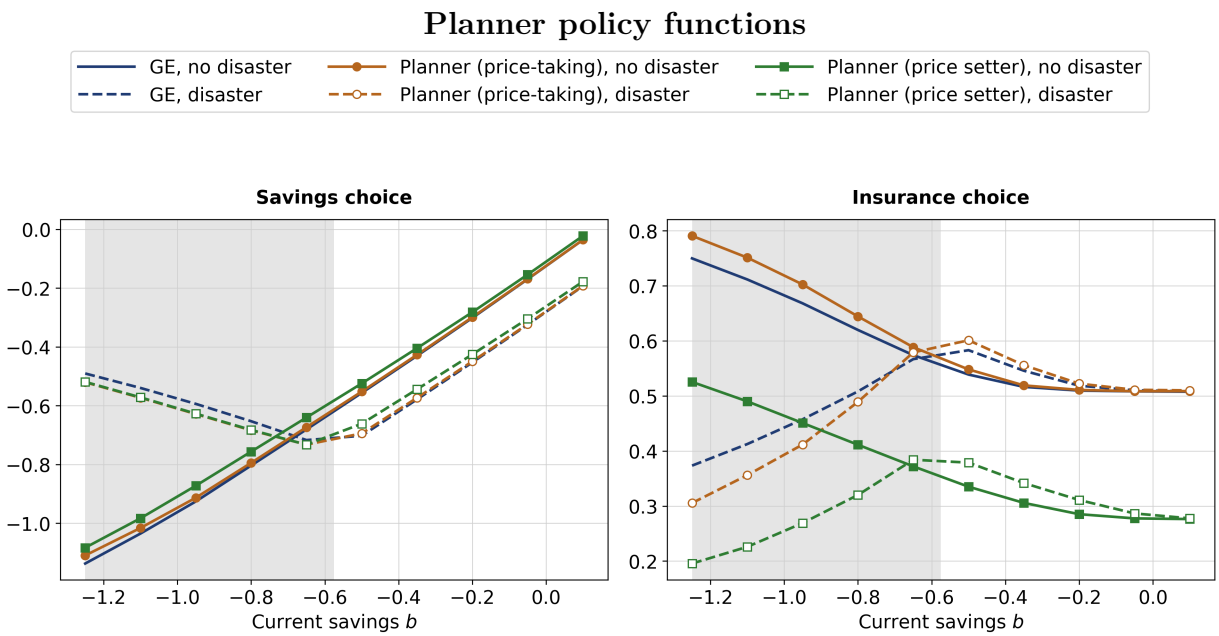



**Figure 9.** Comparison of policy functions in the decentralized (DE) and constrained-efficient (planner) equilibrium across the savings and disaster states. The insurance and net worth states are set to their median and $\kappa$ is set to its highest value.

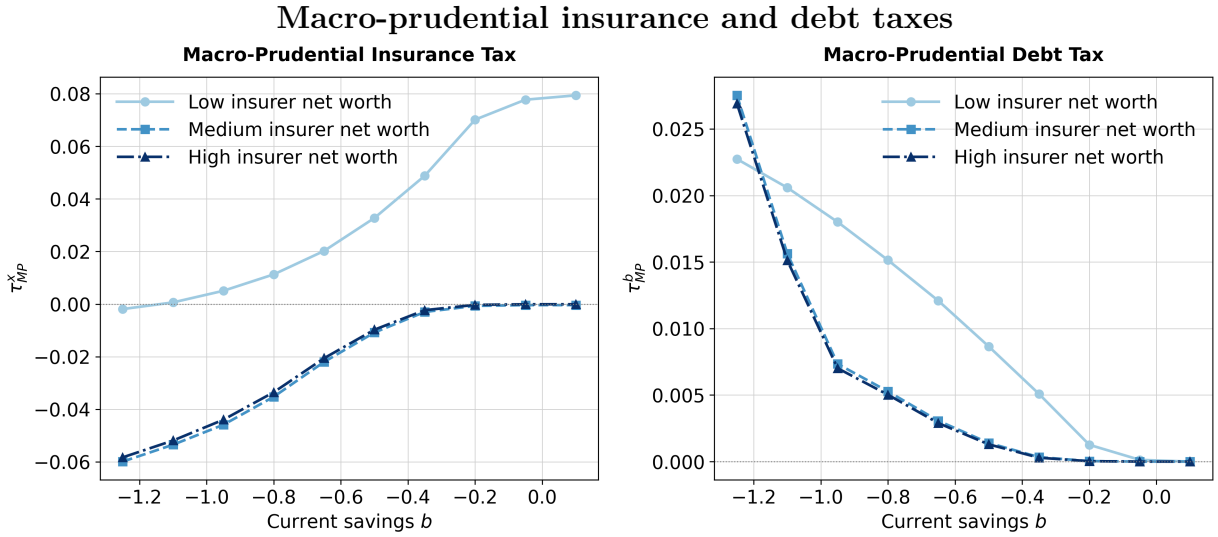



**Figure 10.** Macro-prudential insurance and debt taxes across the savings state for different levels of net worth for the price-taking planner. The insurance state is set to the median, and I consider the no-disaster and high $\kappa$ state.

solid area), the social value of future insurer net worth (green dotted) and corrections to ensure the time-consistency of the optimal insurance policy (orange striped).

The time-consistency correction term reflects the planner internalizing that more insurance in expectation increases future insurer net worth, which coincides with higher house prices tomorrow (see Figure 6). Higher future house price makes that in the next period the

implementability constraint binds less severely. The continuation value of net worth however lowers insurance take-up. When future insurer net worth is expected to be high, so are future insurance quantities. This implies that the possibility of reaching binding collateral constraints in the future is low, which in turn lowers the value of the house as collateral. The planner corrects for this by lowering insurance take-up.

At high levels of savings, households self-insure and incentivizing insurance take-up is not necessary as the probability of reaching binding future constraints, i.e., a state where $\xi_{t+1} > 0$, is zero. When insurer net worth is low, the planner may even tax insurance slightly to prevent a non-linear decline in net worth upon a disaster. At medium levels of savings, the planner optimally subsidizes insurance up to 4% when net worth is lowest. The subsidy is high for low levels of net worth as equilibrium insurance quantities in these states are low. As the marginal benefit of insurance is high when insurance quantities are low, the subsidy is highest. As demand for insurance increases, insurers become more operationally leveraged, steepening the drop in net worth upon a shock. The planner lowers the subsidy when insurers become more fragile, but this effect is modest and the subsidy never changes sign. The decomposition of the insurance subsidy at low savings mimics that of the subsidy at medium savings, although magnitudes grow, with the subsidy reaching 8%.

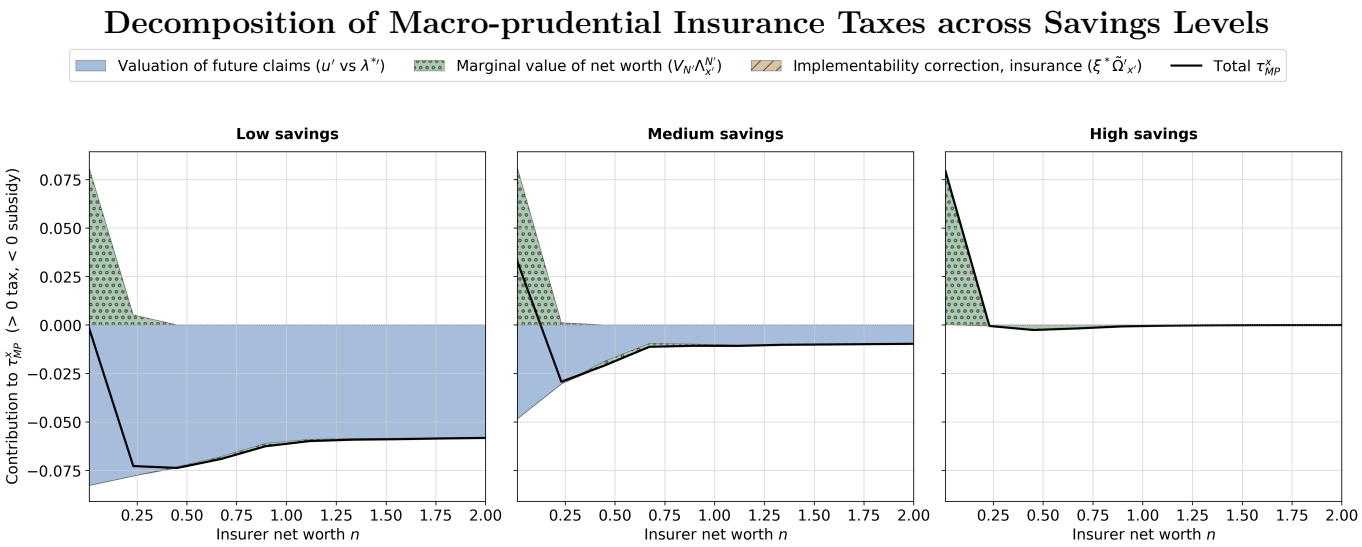



**Figure 11.** Decomposition of the Macro-prudential insurance tax into the household valuation wedge and the insurer net worth wedge. The household valuation wedge captures differences in future marginal utilities when a disaster hits. The insurer net worth wedge corrects for the possibility of future binding capital constraints.

Table 4 presents the results of the simulations in both the planner and decentralized equilibrium. The planner's economy generates welfare gains equal to 0.2% of consumption

equivalents, which is comparable to what Bianchi and Mendoza (2018) find. The average insurance subsidy equals 2.7%, while the debt tax is small and averages 0.5%. The social planner achieves a lower insurance gap by demanding more. While exposing insurers more to disaster risk, insurer fragility, as measured by MPI, is lower. The planner thus achieves a higher insurance penetration while avoiding insurance market crashes by appropriately subsidizing insurance across net worth states.

**Table 4**
Comparison of equilibria and welfare gains. The table reports moments of simulations of 30,000 periods in the decentralized equilibrium and in the economies with a price-taking and a price-setting planner.

| | Decentralized equilibrium | | Planner (price-taking) | | Planner (price-setting) | |
|---|---|---|---|---|---|---|
| | Mean | SD | Mean | SD | Mean | SD |
| *Summary statistics* | | | | | | |
| House price $p^s$ | 1.527 | 0.057 | 1.528 | 0.042 | 1.527 | 0.057 |
| Consumption | 2.955 | 0.065 | 2.956 | 0.049 | 2.965 | 0.069 |
| Marginal Propensity to Insure | 0.332 | 0.364 | 0.256 | 0.347 | 0.639 | 0.034 |
| Insurance premium $p^x$ | 0.058 | 0.001 | 0.059 | 0.002 | 0.021 | 0.001 |
| Insurance gap | 43.6% | – | 41.2% | – | 99.2% | – |
| Collateral constraint binds | 5.3% | – | 5.3% | – | 5.3% | – |
| Insurer constraint binds | 39.6% | – | 29.9% | – | 99.7% | – |
| House price drop at disaster | 16.3% | – | 11.5% | – | 15.6% | – |
| Insurer net worth drop at disaster | 25.6% | – | 24.8% | – | 0.2% | – |
| *Taxes and welfare* | | | | | | |
| Debt tax $\tau^b$ | – | – | 0.47% | 0.06% | 0.56% | 0.10% |
| Insurance tax $\tau^x$ | – | – | −2.65% | 1.89% | −2.45% | 0.73% |
| $\rho_{\tau^b,\tau^x}$ | – | – | 0.334 | – | 0.077 | – |
| Welfare gain | – | – | 0.19% | – | −2.82% | – |

# 8 Conclusion

The availability of homeowners insurance in weather exposed housing markets has become uncertain, posing risks for homeowners and broader financial stability. Existing insurance market regulations prioritize keeping insurance affordable and available, but may ignore how they reduce long-run insurance availability when they undermine the financial conditions of insurers.

This paper develops a dynamic housing model with disaster risk in which intermediary constraints are endogenous. The setup gives rise to a basic intertemporal trade-off: insuring disaster risks lowers household fragility, but increases intermediary fragility. I introduce a social planner who optimizes this intertemporal tradeoff and find that subsidizing insurance is generally optimal to correct for the externality of severe house price crashes upon disaster shocks, though the subsidy lowers when insurers are fragile. Allowing a regulator to directly influence insurance prices lowers welfare when insurer financial constraints and low insurance demand reinforce each other, driving insurance gaps to become persistent. Future research could expand this framework by incorporating other realistic features of insurance markets, such as heterogeneity in climate risk exposure across homeowners, competition among insurers, residual insurers and bail-out guarantees that could affect the regulatory trade-off.

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# 9 Empirics

## 9.1 Empirical Fact 2: Identification strategy

I construct the instrument as the zip-code level weighted average of insurers' exposure to out-of-county damages[17]:

$$Z_{z,t}^{\bar{D}} = \log\left(\sum_{i\in\mathcal{I}_z}\omega_{i,z,t-1}\sum_{\substack{\tilde{c}\in\mathcal{C}\\ d_{c(z),\tilde{c}}>\bar{D}}}\omega^{s}_{\tilde{c},i,t-1}\,\text{WeatherLosses}_{\tilde{c},t-1}\right) \tag{45}$$

where $\mathcal{I}_z$ is the set of insurers active in zip code $z$ and $\mathcal{C}$ is the set of all counties. The function $c(z)$ maps a zip code $z$ to its corresponding county. The instrument considers that disaster damages in the past year affects current insurance prices. While Oh et al. (2021) provide evidence of cross-subsidizing behavior of insurers across states, this analysis focuses on possible cross-subsidization within a state. The instrument can be interpreted as a shock to insurers cross-subsidization behavior. Such behavior is not driven by possible regulatory frictions, but rather updates of fire risk models or financial frictions that drive the insurer to adopt more conservative pricing within the entire state.

As before, $\omega_{i,z,t-1}$ captures the market share of insurer $i$ in zip code $z$, lagged by one year to avoid capturing the effect of contemporaneous disaster losses on market withdrawals. The weight $\omega^{s}_{\tilde{c},i,t-1}$ captures the proportion of total premiums an insurer earns from county $\tilde{c}$, defined as:

$$\omega^{s}_{\tilde{c},i,t-1} = \frac{\text{Premiums}_{i,\tilde{c},t-1}}{\sum_{\tilde{c}'\in\mathcal{C}}\text{Premiums}_{i,\tilde{c}',t-1}}. \tag{46}$$

The variable $\text{WeatherLosses}_{\tilde{c},t-1}$ captures disaster losses as reported by the New York Fed. While these damages may reflect various natural hazards, damages are skewed towards fire hazards in California and hurricanes in Florida.

The validity of the instrument relies on out of county weather losses affecting house prices only through insurance prices and not through local risk perceptions. While regional disaster incidents could affect out-of-county risk perceptions, as long as distant damages, through the insurer's portfolio, are not structurally related to local risk perceptions, the exclusion restriction should be satisfied. The year fixed effects would absorb any changes in aggregate wildfire risk beliefs that could correlate with distant disaster damages and state-wide house

[17] While zip-code damages are available through in CDI, these (i) reflect damages only by reporting insurers and therefore may have the wrong magnitude, (ii) are available from 2018 onward only, and (iii) capture only insured damages. The use of county-level disaster data overcomes these issues.

prices.

To mitigate concerns about spatial correlation in damage that violate the exclusion restriction, losses in counties within $\bar{D}$ miles of each other are excluded, where $d_{c,\tilde{c}}$ is the distance between county $c$ and county $\tilde{c}$. I measure distances between counties based on the centroid of the county. In a small number of cases, one-zip code could be part of more than 1 county. In this case, I allocate the county damages to the zip-code proportional to the share of the population in the zip-code that lives in a given county. The first stage becomes:

$$Y_{z,t} = \beta^z Z^{\bar{D}}_{z,t} + \gamma' X_{z,t} + \eta_z + \tau_t + \epsilon_{z,t} \tag{47}$$

Figure B.1 plots the first-stage coefficient for different distance cutoffs $\bar{D}$. Higher distance cutoffs $\bar{D}$ imply that progressively more distant damages are used to instrument zip-code level insurance prices. The Figure shows that the predictive power of the instrument for insurance cancellations is maximized for intermediate distances around 200 miles. When $\bar{D}$ is small, nearby disasters are included in the instrument. Consequently, local damages that are included as a control and zip-code level fixed effects capture most variation so that the instrument does not yield additional predictive power. When $\bar{D}$ is large, only a small proportion of actual damages to which insurers have exposure is covered, equally reducing its predictive power. The finding that the instrument has predictive power for insurance cancellation after including zip-code fixed effects that absorb time invariant fire risk exposures, suggests that the instrument does not only reflect risk. Relatedly, Figure (B.3) shows that county-level damages have close to zero autocorrelation, suggesting that it is hard to predict where future damages will occur.

In contrast, the first-stage coefficient of the instrument for insurance prices remains small. First-stage F-tests indicate that the instrument does not pass the Yogo-Stock F-value threshold of 10 for most distance cutoffs. For insurance prices, I construct an alternative instrument based on insurers' pricing errors. The instrument $Z^{\text{Resid}}$ captures variation in local insurance prices that reflect deviations from the insurers' pricing model that is trained on other zip-codes.

Table A.1 shows the first stage regression results for the instruments. The positive coefficients on log insurance renewals in column 2 shows that distant climate damages have affected local insurance conditions, revealing spatial spillovers. Columns 3 and 4 show the interaction with the average log of insurer net worth. The negative coefficient on the interaction term suggest that better insurer financial conditions reduce the volume on non-renewals following the same increase in disaster damages. Financial constraints thus affect the mag-

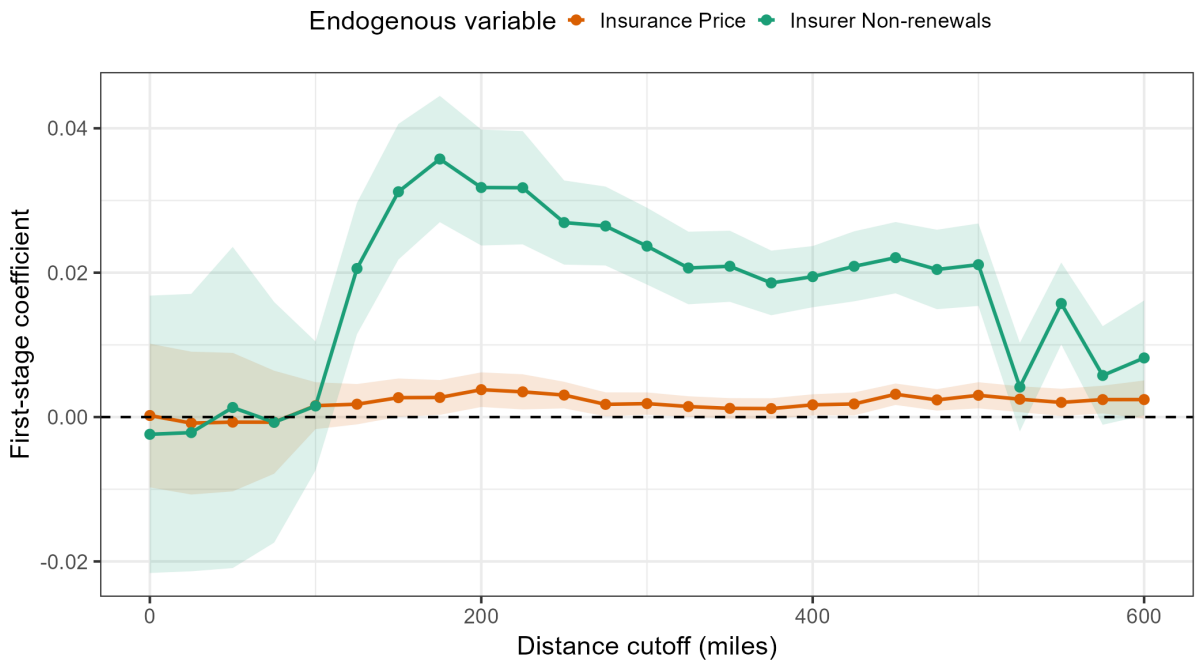



**Figure B.1.** Figure reports the first stage coefficient of the instrumented on endogenous insurance variables. The instruments are constructed with different distance cutoffs $\bar{D}$, separately for the two instrumented variables insurance prices and insurer non-renewals. The zip-code level instrument captures the incumbent insurers' exposure to disaster losses in counties that are at least $\bar{D}$ miles away from the county that belongs to the zip-code, weighted by their zip-code market share.

nitude of spatial spillovers in insurance markets.

**Table A.1**
First stage regression of zip-code level insurance indicators on the instrument, which reflects the logarithm of out of zip-code weather losses weighted by insurers' portfolio weights and local market shares. Standard errors in parentheses are clustered at the zip-code level. Significance codes: ***: 0.01, **: 0.05, *: 0.1

| | Log Ins. Price (1) | Log Ins. Non-renewals (2) | Log Ins. Price (3) | Log Ins. Non-renewals (4) | Δ Log RBC (avg) (5) | Δ Log Net Worth(avg) (6) |
|---|---|---|---|---|---|---|
| $Z^{\text{Resid}}$ | 0.63*** | | 2.0*** | | | |
| | (0.09) | | (0.77) | | | |
| $Z^{200}$ | | 0.03*** | | 0.23** | -0.01*** | -0.003*** |
| | | (0.004) | | (0.09) | (0.001) | (0.0004) |
| Log Net Worth (avg) | | | 0.06*** | 0.14 | | |
| | | | (0.01) | (0.09) | | |
| $Z^{\text{Resid}}\times$ Log Net Worth (avg) | | | −0.10* | | | |
| | | | (0.06) | | | |
| $Z^{200}\times$ Log Net Worth (avg) | | | | −0.01** | | |
| | | | | (0.006) | | |
| Zip-code FE | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes |
| F-test | 52.2 | 51.7 | 153.3 | 81.9 | 38.5 | 27.1 |
| Observations | 14,469 | 8,029 | 14,469 | 8,029 | 14,935 | 15,046 |
| $R^2$ | 0.92 | 0.96 | 0.93 | 0.96 | 0.29 | 0.71 |
| Within $R^2$ | 0.23 | 0.05 | 0.29 | 0.05 | 0.02 | 0.02 |

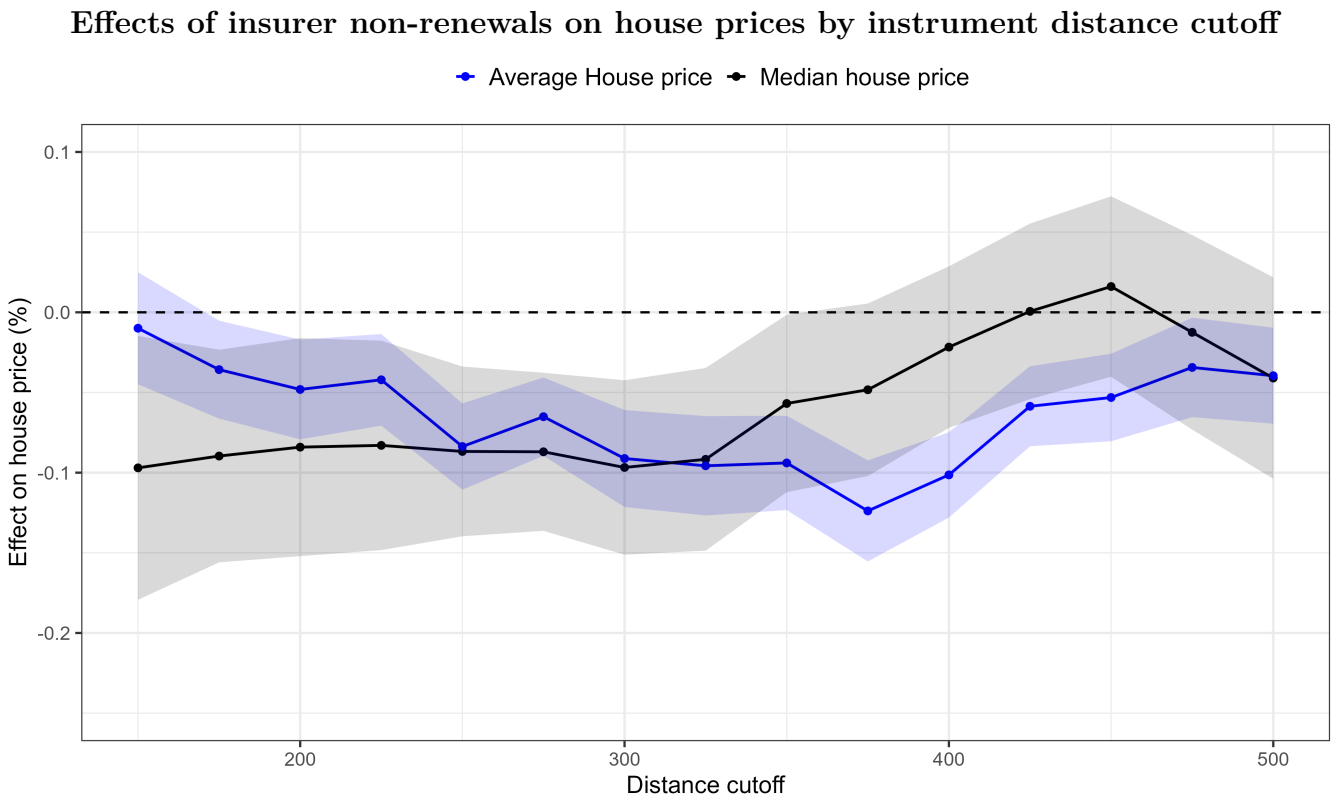



**Figure B.2.** Second stage coefficient estimates in a regression of log house prices (median or average) on the log of insurer-driven non-renewals at the zip-code level, which is instrumented with out-of-zip-code weather damages weighted by the incumbent insurers' out-of-zip-code portfolio shares. The regression is ran for different distance cutoffs, that determine which damages are included in the instrument. Standard errors are clustered at the zip-code level.

**Kernel Density plot of auto-correlation and volatility in disaster damages**

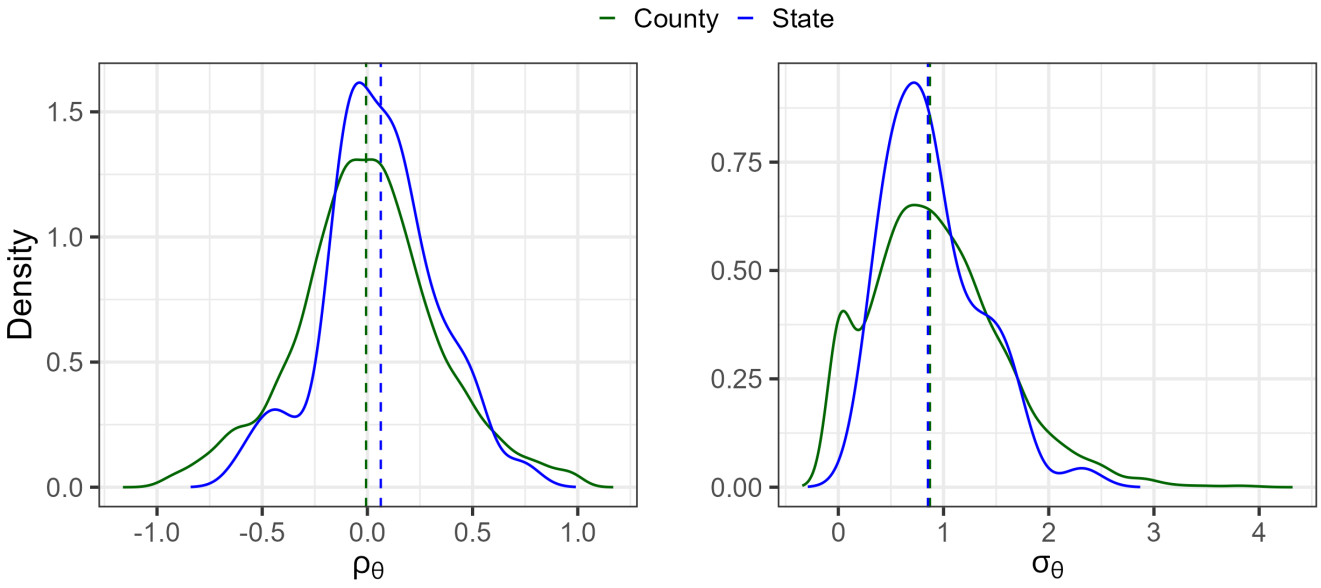



**Figure B.3.** Autocorrelation (left) and volatility (right) is computed at both the state and county level in the period 1994 - 2024. Dashed vertical line shows median autocorrelation and volatility coefficient by state and county. Disaster severity is calculated as dollar damages divided by total income at the county or state level. All types of disasters are included.

**Table A.2**
Empirical estimates of marginal propensity to insure (panel A) and exit (panel B). Standard errors in parentheses are clustered at the insurer level. Significance codes: ***: 0.01, **: 0.05, *: 0.1

| Panel A | | | | |
|---|---|---|---|---|
| | Log Exposure | | $\Delta$ Log Exposure | |
| Model: | (1) | (2) | (3) | (4) |
| Log Net Worth | 0.27*** | 0.75*** | | |
| | (0.10) | (0.15) | | |
| $\Delta$ Log Net Worth | | | 0.62*** | 0.08 |
| | | | (0.17) | (0.16) |
| Zip-code×Year FE | Yes | Yes | Yes | Yes |
| Insurer FE | No | Yes | No | Yes |
| Observations | 724,911 | 724,911 | 625,809 | 625,809 |
| $R^2$ | 0.24 | 0.61 | 0.06 | 0.24 |
| Within $R^2$ | 0.10 | 0.03 | 0.03 | 0.0004 |
| Panel B | | | | |
| | $\text{Exit}_{t+1}$ | | | |
| Model: | (1) | (2) | (3) | (4) |
| Log Net Worth | -0.009** | -0.09*** | | |
| | (0.004) | (0.02) | | |
| $\Delta$ Log Net Worth | | | -0.10** | 0.03 |
| | | | (0.05) | (0.05) |
| Zip-code×Year FE | Yes | Yes | Yes | Yes |
| Insurer FE | No | Yes | No | Yes |
| Observations | 665,018 | 665,018 | 665,018 | 665,018 |
| $R^2$ | 0.09 | 0.21 | 0.09 | 0.20 |
| Within $R^2$ | 0.008 | 0.02 | 0.003 | 0.0002 |

**Table A.3**
Heterogeneity in marginal propensity to insure. Standard errors in parentheses are clustered at the insurer level. Significance codes: ***: 0.01, **: 0.05, *: 0.1

| | Δ Log Exposure | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| Δ Log Net Worth | 1.9** | -0.74 | 0.69*** | 1.3 |
| | (0.79) | (0.55) | (0.18) | (0.91) |
| Δ Log Net Worth × Log RBC Ratio | -0.19* | | | |
| | (0.12) | | | |
| Δ Log Net Worth × Leverage | | 2.2** | | |
| | | (0.87) | | |
| Δ Log Net Worth × Market Share | | | -0.03*** | |
| | | | (0.008) | |
| Δ Log Net Worth × Log Total Assets | | | | -0.05 |
| | | | | (0.06) |
| Zip-code×Year FE | Yes | Yes | Yes | Yes |
| Insurer FE | No | No | No | No |
| Observations | 560,156 | 625,809 | 625,809 | 625,809 |
| $R^2$ | 0.07 | 0.07 | 0.06 | 0.06 |
| Within $R^2$ | 0.04 | 0.04 | 0.03 | 0.03 |

**Table A.4**
The role of insurer financial conditions in insurance prices and non-renewals. Standard errors in parentheses are clustered at the zip-code level. Significance codes: ***: 0.01, **: 0.05, *: 0.1

| | Δ Log Ins. Price | | | | Log Non-renewals | | | |
|---|---|---|---|---|---|---|---|---|
| | OLS | IV | OLS | IV | OLS | IV | OLS | IV |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Δ Log Net Worth (avg) | -0.19** | -2.9** | | | -0.35** | -3.8* | | |
| | (0.09) | (1.2) | | | (0.16) | (2.1) | | |
| Δ Log RBC (avg) | | | 0.004 | -0.45*** | | | -0.35*** | -0.57** |
| | | | (0.03) | (0.17) | | | (0.07) | (0.28) |
| Controls | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Zip-Code FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Year FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Observations | 15,046 | 15,046 | 14,935 | 14,935 | 8,043 | 8,029 | 8,030 | 8,018 |
| $R^2$ | 0.16 | 0.16 | 0.18 | 0.18 | 0.96 | 0.96 | 0.96 | 0.96 |
| Within $R^2$ | 0.01 | 0.01 | 0.01 | 0.01 | 0.04 | 0.05 | 0.05 | 0.05 |

**Table A.5**
OLS Regression of house prices on the share of the residual insurer in California (FAIR Plan). Standard errors in parentheses are clustered at the zip-code level. Significance codes: ***: 0.01, **: 0.05, *: 0.1

| | Log(Median HP) | Log(Mean HP) |
|---|---|---|
| | (1) | (2) |
| FAIR Plan Share | -0.41*** | -0.25*** |
| | (0.07) | (0.04) |
| Controls | Yes | Yes |
| Zip-code FE | Yes | Yes |
| Year FE | Yes | Yes |
| Observations | 12,768 | 11,753 |
| $R^2$ | 0.96 | 0.99 |
| Within $R^2$ | 0.09 | 0.16 |

# 10 Model

## 10.1 Household optimality conditions - decentralized equilibrium

The Lagrangian for the household is:

$$\begin{aligned}\mathcal{L} = \mathbb{E}_t \sum_{k=0}^{\infty} \beta^k \Bigg\{ & \frac{1}{1-\psi} \left(c_{t+k} + \nu_s g(s_{t+k})\right)^{1-\psi} \\ & + \lambda_{t+k} \Bigg[ b_{t+k} + h_{t+k} w_{t+k} + p^s_{t+k} s_{t+k} + \mathbf{1}_{t+k} \phi_{t+k} s_{t+k} (x_{t+k} - 1) + l_{t+k} - f(l_{t+k}) \\ & \quad - c_{t+k} - \frac{b_{t+1+k}}{R_{t+k}} - s_{t+1+k} p^s_{t+k} - p^x_{t+k} x_{t+1+k} s_{t+1+k} p^r \Bigg] \\ & + \mu_{t+k} \Bigg[ \kappa_{t+k}(x_{t+1+k}) \times \left( p^s_{t+k} s_{t+k} \left(1 - \mathbf{1}_{t+k}\phi_{t+k}\right) + \mathbf{1}_{t+k}\phi_{t+k} s_{t+k} x_{t+k} p^r - l_{t+k} \right) + \frac{b_{t+1+k}}{R_{t+k}} \Bigg] \Bigg\}\end{aligned}$$

Define the composite $\mathcal{X}_t = c_t + \nu_s g(s_t)$ and noting that $\lambda_t = \mathcal{X}_t^{-\psi} = U'(t)$, the optimality condition for borrowing is:

$$U'(t) = \beta R_t \mathbb{E}_t \left[U'(t+1)\right] + \mu_t \tag{48}$$

The first-order condition for insurance demand $x_{t+1}$ yields the following equation:

$$-\lambda_t p^x_t s_{t+1} + \beta \mathbb{E}_t \left[\mathbf{1}_{t+1} \phi_{t+1} p^r \left(\lambda_{t+1} + \mu_{t+1}\right)\right] s_{t+1} = 0 \tag{49}$$

Rearranging gives:

$$p^x_t \lambda_t = \beta \mathbb{E}_t \left[\mathbf{1}_{t+1} \phi_{t+1} p^r (\lambda_{t+1} + \mu_{t+1} \kappa_{t+1})\right] \tag{50}$$

The first-order condition for housing $s_{t+1}$ yields:

$$\begin{aligned}\lambda_t (p^s_t + p^x_t x_{t+1}) = \beta \mathbb{E}_t \Bigg[ & \lambda_{t+1} \Big\{ \nu_s g'(s_{t+1}) + p^s_{t+1} + \mathbf{1}_{t+1} \phi_{t+1} x_{t+1} p^r \Big\} \\ & + \mu_{t+1} \kappa_{t+1} (p^s_{t+1} (1 - \mathbf{1}_{t+1} \phi_{t+1}) + \mathbf{1}_{t+1} \phi_{t+1} x_{t+1} p^r) \Bigg]\end{aligned} \tag{51}$$

Substituting the insurance condition into the housing condition gives:

$$\lambda_t p^s_t = \beta \mathbb{E}_t \left[ \lambda_{t+1} \left\{ \nu_s g'(s_{t+1}) + p^s_{t+1} \right\} + \mu_{t+1} \kappa_{t+1} (p^s_{t+1} (1 - \mathbf{1}_{t+1} \phi_{t+1})) \right] \tag{52}$$

The first order conditions for home equity redraws $l_t$ gives:

$$1 = \theta l_t + \frac{\mu_t \kappa_t}{\lambda_t} \tag{53}$$

**Derivation house price** If the collateral constraint does not bind, i.e., $\mu_t = 0$, then equation (13) yields:

$$p_t^s = \mathbb{E}_t \left[ m_{t,t+1} \left( \nu_s g'(s_{t+1}) + q_{t+1}^s \right) \right] \tag{54}$$

Iterating one-period forward gives:

$$q_{t+1}^s = \mathbb{E}_{t+1} \left[ m_{t+1,t+2} \left( \nu_s g'(s_{t+2}) + q_{t+2}^s \right) \right] \tag{55}$$

Substituting back and using the law of iterated expectations yields:

$$p_t^s = \mathbb{E}_t \left[ m_{t,t+1} \nu_s g'(s_{t+1}) \right] + \mathbb{E}_t \left[ m_{t,t+1} m_{t+1,t+2} \nu_s g'(s_{t+2}) \right] + \mathbb{E}_t \left[ m_{t,t+1} m_{t+1,t+2} q_{t+2}^s \right] \tag{56}$$

Iterating this forward and using the transversality condition, one can express the value of the housing investment today as the expected present value of housing gains (marginal housing utility) discounted with $R_{t+1}^s$:

$$p_t^s = \mathbb{E}_t \left[ \sum_{j=0}^{\infty} \left( \prod_{i=0}^{j} \mathbb{E}_{t+i} R_{t+1+i}^s \right)^{-1} \nu_s g'(s_{t+j+1}) \right] \tag{57}$$

**Excess housing return** We can combine equations (10) and (13) to derive the expected excess return on housing $R^{hp} = \mathbb{E}_t \left[ R_{t+1}^s - R_t \right]$. Note that (10) can be written as:

$$1 = \mathbb{E}_t \left[ m_{t+1} \right] R_t + \frac{\mu_t}{\lambda_t} \tag{58}$$

Similarly, (13) can be written as:

$$1 = \mathbb{E}_t \left[ m_{t+1} \frac{\nu_s g'(s_{t+1}) + q_{t+1}^s}{p_t^s} \right] + \beta \mathbb{E}_t \left[ \frac{\mu_{t+1} \kappa_{t+1} q_{t+1}^s}{\lambda_t p_t^s} \right] \tag{59}$$

$$1 = \mathbb{E}_t \left[ m_{t+1} R_{t+1}^s \right] + \mathbb{E}_t \left[ m_{t+1} \Upsilon_{t+1} \right] \tag{60}$$

Subtracting both equations yields:

$$\mathbb{E}_t\left[m_{t+1}(R^s_{t+1}-R_t)\right]=\frac{\mu_t}{\lambda_t}-\mathbb{E}_t\left[m_{t+1}\Upsilon_{t+1}\right] \tag{61}$$

$$R^{hp}_t=\mathbb{E}_t\left[R^s_{t+1}-R_t\right]=\underbrace{\frac{\mu_t}{\lambda_t\,\mathbb{E}_t\left[m_{t+1}\right]}}_{\text{Liquidity premium}}-\underbrace{\frac{\mathbb{E}_t\left[m_{t+1}\,\Upsilon_{t+1}\right]}{\mathbb{E}_t\left[m_{t+1}\right]}}_{\text{Collateral effect}}-\underbrace{\frac{\mathrm{Cov}_t\left(m_{t+1},R^s_{t+1}\right)}{\mathbb{E}_t\left[m_{t+1}\right]}}_{\text{Risk premium}} \tag{62}$$

where

$$\Upsilon_{t+1}=\frac{\mu_{t+1}\kappa_{t+1}q^s_{t+1}}{\lambda_{t+1}p^s_t} \tag{63}$$

## 10.2 Goods market clearing

Start from the household budget constraint:

$$c=b-b'/R-p^xx'+hw-p^s\mathbf{1}\phi+\mathbf{1}\phi x+\Pi \tag{64}$$

Substitute the final good firm's sectoral constraint $y=wh+q^aR^k\Phi(a)$ gives:

$$c=y+b-b'/R-p^xx'-q^aR^k\Phi(a)-p^s\mathbf{1}\phi+\mathbf{1}\phi x+\Pi \tag{65}$$

Substituting $\Pi=n_t-N_t$ and noting that $n_t=a_t(1+R^k)-\mathbf{1}\phi x$ gives:

$$c=y+b-b'/R-p^xx'-q^aR^k\Phi(a)-p^s\mathbf{1}\phi+a_t(1+R^k)-N_t \tag{66}$$

Substituting the insurer balance sheet constraint $N_t=a-p^xx'$ gives:

$$c=y+b-b'/R-p^s\mathbf{1}\phi+R^k(a_t-q^a\Phi(a)) \tag{67}$$

Note that the perfect competition argument results in the last term dropping out:

$$c=y+b-b'/R-p^s\mathbf{1}\phi \tag{68}$$

## 10.3 Planner optimality conditions

The planner's problem in recursive form can be formulated as below. The planner's state is $(b, x, n, \mathcal{S})$, where $N$ is aggregate insurer net worth after exit and entry.

$$
\begin{aligned}
\mathcal{L} = & \, u(c,1) + \beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}}\Big[V(b', x', n', \mathcal{S}')\Big] \\
& + \lambda^* \left[ b + w + \mathbf{1}\phi\left[x p^r - 1\right] + l - f(l) - c - \frac{b'}{R} - p^x x' \right] \\
& + \mu^* \left[ \kappa\Big( p^s\big(1 - \mathbf{1}\phi\big) + \mathbf{1}\phi x p^r - l \Big) + \frac{b'}{R} \right] \\
& + \xi^* \left[ \beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ u'\Big(c'(b', x', n', \mathcal{S}'),\, 1\Big) \Big\{ \nu_s g'(1) + p^{s'}(b', x', n', \mathcal{S}') \Big\} \right.\right. \\
& \left.\left. + \mu'(b', x', n', \mathcal{S}')\, \kappa' Q'(b', x', n', \mathcal{S}') \right] - u'(c,1)\, p^s \right] \\
& + \Psi^* \Big[ \mathcal{X}(p^x, n, \mathcal{S}) - x' \Big]
\end{aligned}
\tag{69}
$$

where $\mathcal{X}(p^x, n, \mathcal{S})$ is the optimal supply decision of the insurer given the insurance price and its net worth, and the value of the house net of disaster shocks equals $Q'(b', x', n', \mathcal{S}') = p^{s'}(b', x', n', \mathcal{S}')(1 - \mathbf{1}'\phi')$.

**Preliminaries** Insurers invest premiums through $a = N + p^x x'$. Gross net worth before exit and entry is $n' = a(1 + R'^k) - \mathbf{1}\phi\, x'\, p^r + \zeta^1 x'^{\zeta^2}$. The derivative of next-period net worth with respect to the insurance choice, insurance price and net worth today are:

$$
\begin{aligned}
\Lambda^{N'}_{x'} &= \frac{\partial N'}{\partial x'} = p^x(1 + R'^k) - \Big(\mathbf{1}'\phi' p^r + \zeta^1\zeta^2 x'^{\zeta^2 - 1}\Big), \\
\Lambda^{N'}_{p^x} &= \frac{\partial N'}{\partial p^x} = (1 + R'^k)\, x', \\
\Lambda^{N'}_{N} &= \frac{\partial N'}{\partial N} = (1 + R'^k)
\end{aligned}
\tag{70}
$$

The envelope conditions are as follows:

$$V_b = \lambda^* \tag{71}$$

$$V_x = \left(\lambda^* + \mu^* \kappa\right) \mathbf{1} \phi \, p^r \tag{72}$$

$$V_N = \Psi^* \mathcal{X}_N(p^x, n, \mathcal{S}) + \beta \, \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ \Lambda_N^{N'} \left( V_{N'} + \xi^* \tilde{\Omega}'_{N'} \right) \right] \tag{73}$$

where $\mathcal{X}_N$ is the marginal propensity to insure and $\tilde{\Omega}'_{N'}$ is the derivative of the right-hand side of the implementability constraint with respect to next-period net worth:

$$\begin{aligned} \tilde{\Omega}'_{N'} = {} & u''(c'(b', x', n', \mathcal{S}'), 1) \left( \nu_s g'(1) + p^{s'}(b', x', n', \mathcal{S}') \right) c'_{N'}(b', x', n', \mathcal{S}') \\ & + u'(c'(b', x', n', \mathcal{S}'), 1) \, p^{s'}_{N'}(b', x', n', \mathcal{S}') \\ & + \mu'_{N'}(b', x', n', \mathcal{S}') \, \kappa' Q'(b', x', n', \mathcal{S}') + \mu'(b', x', n', \mathcal{S}') \, \kappa' Q'_{N'}(b', x', n', \mathcal{S}'). \end{aligned} \tag{74}$$

Taking the first-order condition with respect to $c$ yields:

$$u'(c, 1) = \lambda^* + u''(c, 1) \, \xi^* p^s \tag{75}$$

The first-order condition for $b'$ yields:

$$\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \lambda'^* - \frac{\lambda^*}{R} + \frac{\mu^*}{R} + \beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ \xi^* \tilde{\Omega}'_{b'} \right] = 0 \tag{76}$$

where

$$\begin{aligned} \tilde{\Omega}'_{b'} = {} & u''(c'(b', x', n', \mathcal{S}'), 1) \left( \nu_s g'(1) + p^{s'}(b', x', n', \mathcal{S}') \right) c'_{b'}(b', x', n', \mathcal{S}') \\ & + u'(c'(b', x', n', \mathcal{S}'), 1) \, p^{s'}_{b'}(b', x', n', \mathcal{S}') \\ & + \mu'_{b'}(b', x', n', \mathcal{S}') \, \kappa' Q'(b', x', n', \mathcal{S}') + \mu'(b', x', n', \mathcal{S}') \, \kappa' Q'_{b'}(b', x', n', \mathcal{S}'). \end{aligned} \tag{77}$$

Multiplying by $R$ and using the FOC for consumption gives:

$$u'(c, 1) = \beta R \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left( u'(c', 1) - u''(c', 1) \, \xi'^* p^{s'} + \xi^* \tilde{\Omega}'_{b'} \right) + u''(c, 1) \, \xi^* p^s + \mu^* \tag{78}$$

The first-order condition for $x'$ yields:

$$-\lambda^* p^x + \beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ (\lambda'^* + \mu'^* \kappa') \mathbf{1}' \phi' p^r + \lambda'^* \, \Pi'_{x'} + V_{N'} \, \Lambda^{N'}_{x'} + \xi^* \tilde{\Omega}'_{x'} \right] - \Psi^* = 0 \tag{79}$$

where

$$\begin{aligned}\tilde{\Omega}'_{x'} = {} & u''(c'(b',x',n',\mathcal{S}'),1)\left(\nu_s g'(1) + p^{s'}(b',x',n',\mathcal{S}')\right)\left(c'_{x'}(b',x',n',\mathcal{S}') + c'_{N'}(b',x',n',\mathcal{S}')\,\Lambda^{N'}_{x'}\right) \\ & + u'(c'(b',x',n',\mathcal{S}'),1)\left(p^{s'}_{x'}(b',x',n',\mathcal{S}') + p^{s'}_{N'}(b',x',n',\mathcal{S}')\,\Lambda^{N'}_{x'}\right) \\ & + \left(\mu'_{x'} + \mu'_{N'}\,\Lambda^{N'}_{x'}\right)(b',x',n',\mathcal{S}')\,\kappa' Q'(b',x',n',\mathcal{S}') \\ & + \mu'(b',x',n',\mathcal{S}')\,\kappa'\left(Q'_{x'}(b',x',n',\mathcal{S}') + Q'_{N'}(b',x',n',\mathcal{S}')\,\Lambda^{N'}_{x'}\right).\end{aligned} \tag{80}$$

Rewriting gives:

$$p^x\lambda^* + \Psi^* = \beta\mathbb{E}_{\mathcal{S}'|\mathcal{S}}\left[(\lambda'^* + \mu'^*\kappa')\mathbf{1}'\phi' p^r + V_{N'}\,\Lambda^{N'}_{x'} + \xi^*\tilde{\Omega}'_{x'}\right] \tag{81}$$

The first-order condition for the house price $p^s$ gives:

$$\mu^*\kappa(1 - \mathbf{1}\phi) - \xi^* u'(c,1) = 0 \tag{82}$$

Solving for $\xi^*$ gives:

$$\xi^* = \frac{\mu^*\kappa(1-\mathbf{1}\phi)}{u'(c,1)} \tag{83}$$

The first-order condition for the insurance price $p^x$ gives:

$$\lambda^* x' = \Psi^* \mathcal{X}_{p^x}(p^x,n,\mathcal{S}) + \beta\mathbb{E}_{\mathcal{S}'|\mathcal{S}}\left[\xi^*\tilde{\Omega}'_{p^x} + V_{N'}\,\Lambda^{N'}_{p^x}\right] \tag{84}$$

where

$$\tilde{\Omega}'_{p^x} = \Lambda^{N'}_{p^x}\,\tilde{\Omega}'_{N'}. \tag{85}$$

Solving for $\Psi^*$ yields:

$$\Psi^* = \lambda^* x' \mathcal{X}^{-1}_{p^x} - \beta\mathbb{E}_{\mathcal{S}'|\mathcal{S}}\left[\xi^*\tilde{\Omega}'_{p^x} + V_{N'}\,\Lambda^{N'}_{p^x}\right]\mathcal{X}^{-1}_{p^x} \tag{86}$$

Substituting this equation in (81) gives:

$$\begin{aligned}\lambda^*(p^x + x'\mathcal{X}^{-1}_{p^x}) = \beta\mathbb{E}_{\mathcal{S}'|\mathcal{S}}\Big[ & (\lambda'^* + \mu'^*\kappa')\mathbf{1}'\phi' p^r + V_{N'}\left(\Lambda^{N'}_{x'} + \Lambda^{N'}_{p^x}\mathcal{X}^{-1}_{p^x}\right) \\ & + \xi^*\left(\tilde{\Omega}'_{x'} + \tilde{\Omega}'_{p^x}\mathcal{X}^{-1}_{p^x}\right)\Big]\end{aligned} \tag{87}$$

This condition jointly determines the insurance price and quantity, with $x' = \mathcal{X}(p^x,n,\mathcal{S})$. The multiplier $\Psi^*$ then follows from (81), and $V_N$ is updated using (73).

**Deriving the price elasticity of insurer supply $\mathcal{X}_{p^x}$** Define $\nu_A = \mathbb{E}\Big[\Omega'(1+R'^k)\Big]$, $\nu_x = \mathbb{E}\Big[\Omega'\,\mathbf{1}'\phi'\Big]\,p^r$ and $m = \mathbb{E}\Big[\Omega'\Big]$ as the marginal value of assets, marginal value of insurance and the insurer's augmented discount factor. The insurer's value can then be written as

$$V(x') = \nu_A N + \Theta\, x' - m\,\zeta^1 x'^{\zeta^2}, \tag{88}$$

where

$$\Theta = \nu_A\,(p^x + \nu_{ex}) - \nu_x \tag{89}$$

*Unconstrained region.* Solving (25) for the insurance choice gives:

$$x'^{\,unc} = \left(\frac{\Theta}{m\,\zeta^1\zeta^2}\right)^{\frac{1}{\zeta^2-1}}. \tag{90}$$

Differentiating with respect to $p^x$, using $\partial\Theta/\partial p^x = \nu_A$, and substituting the first-order condition:

$$\mathcal{X}^{unc}_{p^x} = \frac{\nu_A}{m\,\zeta^1\zeta^2(\zeta^2-1)\,x'^{\zeta^2-2}} = \frac{x'}{\zeta^2-1}\,\frac{\nu_A}{\Theta} \tag{91}$$

*Constrained region.* When the regulatory constraint binds, $\zeta p^r x' = V(x')$, supply solves

$$G(x', p^x, N) = m\,\zeta^1 x'^{\zeta^2} + \Big(\zeta p^r - \Theta\Big)\,x' - \nu_A N = 0. \tag{92}$$

Its partial derivatives are

$$G_{x'} = \zeta p^r - \Theta + m\,\zeta^1\zeta^2\,x'^{\zeta^2-1} \tag{93}$$

$$G_{p^x} = -\nu_A\,x' \tag{94}$$

By the implicit function theorem,

$$\mathcal{X}^{con}_{p^x} = -\frac{G_{p^x}}{G_{x'}} = \frac{\nu_A\,x'}{\zeta p^r - \Theta + m\,\zeta^1\zeta^2\,x'^{\zeta^2-1}} = x'\,\mathcal{X}^{con}_N \tag{95}$$

In the constrained region, a unit increase in $p^x$ replenishes assets by $x'$, allowing to expand the supply of insurance. The price elasticity of insurance supply is thus $x'$ times marginal propensity to insure.

## 10.4 Optimal taxation

$$p_t^x u'(c,1)(1+\tau^x) = \beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ (\mathbf{1}'\phi' p^r (u'(c',1) + \mu'\kappa') \right] \tag{96}$$

One has to find $\tau^x$ that yields the first order condition of the social planner.

This approach proceeds by inserting as $p_t^x$ the valuation of insurance by the planner. First rewrite the insurance price as the ratio of expected future marginal benefits to marginal costs today:

$$p_t^x(1+\tau^x) = \frac{\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ MB_{t+1}^{DE} \right]}{MC_t^{DE}} \tag{97}$$

Inserting for $p_t^x$ the insurance valuation by the planner yields:

$$\frac{\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ MB_{t+1}^{PL} \right]}{MC_t^{PL}}(1+\tau^x) = \frac{\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ MB_{t+1}^{DE} \right]}{MC_t^{DE}} \tag{98}$$

Solving for the tax yields:

$$\tau^x = \frac{\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ MB_{t+1}^{DE} \right]}{MC_t^{DE}} \frac{MC_t^{PL}}{\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ MB_{t+1}^{PL} \right]} - 1 \tag{99}$$

The insurance tax thus the ratio of the marginal benefit of insurance in the unconstrained's economy to the marginal benefit of insurance in the planner's economy corrected for differences in the marginal cost of insurance today. Equivalently, the tax equals the ratio of the insurance price in the DE to the insurance price in the planner's economy minus 1. Whenever the implementability and feasibility constraints today do not bind, one can derive the macro-prudential (MP) tax on insurance:

$$\tau_{MP}^x = \frac{\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ (u'(c',1) + \mu'\kappa') \mathbf{1}'\phi' p^r \right]}{\beta \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left[ (\lambda'^* + \mu'^* \kappa') \mathbf{1}' \, \phi' p^r + \lambda'^* \, \Pi'_{x'} + V_{N'} \, \Lambda_{x'}^{N'} \right]} - 1 \tag{100}$$

Using similar steps, one can derive the optimal tax on borrowing as

$$\tau^b = \frac{\beta R \mathbb{E}_{\mathcal{S}'|\mathcal{S}} \left( u'(c',1) - \xi^{*\prime} u''(c',1) p'^s + \xi^* \hat{\Omega}'_{b'} \right) + u''(c,1) \xi^* p^s}{\beta R \mathbb{E}_{\mathcal{S}'|\mathcal{S}} u'(c',1)} - 1 \tag{101}$$

Note that when the tax is implemented, the allocations of the planner and in the DE converge, so that $\mu = \mu^*$.

## 10.5 Computational Algorithm - decentralized equilibrium

The algorithm follows the approach of Mendoza and Villalvazo (2020) and solves the model's equilibrium conditions in recursive form. Denote $\mathcal{S} = \{\mathbf{1}, \theta, R\}$ as the triplet of exogenous shocks at date $t$, which includes the disaster indicator, disaster severity and the exogenous interest rate. The two endogenous state variables for the household are borrowing $b$ and insurance $x$. The state variable for the insurer is net worth, which also enter the household problem. The recursive equilibrium is defined as a set of recursive functions for allocations $\Big[c(b,x,n,\mathcal{S}),\ h(b,x,n,\mathcal{S}),\ b'(b,x,n,\mathcal{S}),\ x'(b,x,n,\mathcal{S})\Big]$, prices $\Big[p^x(b,x,n,\mathcal{S}),\ p^s(b,x,n,\mathcal{S})\Big]$, and multiplier $\Big[\mu(b,x,n,\mathcal{S})\Big]$.

- **Step 0**: Start iteration $j$ with an initial conjecture of the pricing functions $\hat{p}^x_j(b,x,n,\mathcal{S})$, $\hat{p}^s_j(b,x,n,\mathcal{S})$, decision rule for bonds $\hat{b}'_j(b,x,n,\mathcal{S}) = b$, and the multiplier $\hat{\mu}_j(b,x,n,\mathcal{S}) = 0$. Iterate on the insurer optimality condition (**??**) until the return on assets $R^k$, marginal values $\nu^a$, $\nu^x$ and allocations $\psi$, $\Psi$ converge. Compute insurance $x'_j(b,x,n,\mathcal{S})$ and asset quantities $a'_j(b,x,n,\mathcal{S})$ using the insurer leverage conditions. Compute equilibrium wages using insurer investments, and the household's and non-financial firms' labor optimality conditions.

- **Step 1**: Compute $l_j(b,x,n,\mathcal{S})$ using the first-order condition for the within-period loan and $\hat{\mu}_j$ and $\hat{c}_j$. Using $x'_j(b,x,n,\mathcal{S})$ and pricing functions, consumption then follows from the resource constraint:

$$c_j(b,x,n,\mathcal{S}) = b + w(b,x,n,\mathcal{S}) + \hat{p}^s_j(b,x,n,\mathcal{S})\Big(1 - \mathbf{1}\,\phi\Big) + \mathbf{1}\,\phi\,x + l_j(b,x,n,\mathcal{S}) - f(l_j(b,x,n,\mathcal{S})) - \frac{\hat{b}'_j(b,x,n,\mathcal{S})}{R} - \hat{p}^s_j(b,x,n,\mathcal{S}) - \hat{p}^x_j(b,x,n,\mathcal{S})\,x'(b,x,n,\mathcal{S}) \tag{102}$$

  Note that housing drops out given the market clearing condition $s = 1$.

- **Step 2**: Assume the collateral constraint does not bind. Solve for $c_{j+1}(b,x,n,\mathcal{S})$ using the Euler equation:

$$c_{j+1}(b,x,n,\mathcal{S}) = \left\{\beta R \mathbb{E}_t\left[\Big(c_j\Big(\hat{b}'_j(b,x,n,\mathcal{S}),\ x'(b,x,n,\mathcal{S}),\ \mathcal{S}'\Big) + \nu_s g(1)\Big)^{-\psi}\right]\right\}^{-\frac{1}{\psi}} - \nu_s g(1) \tag{103}$$

  When computing $c_j(\cdot)$ and $h_j(\cdot)$, use interpolation if $\hat{b}'_j(b,x,n,\mathcal{S})$ and $x'(b,x,\mathcal{S})$ are intermediate points that are not on the grid, and extrapolation if $x'(b,x,n,\mathcal{S})$ or $\hat{b}'_j(b,x,n,\mathcal{S})$ exceed the boundaries of the grid.

- **Step 3**: Solve for $b'_{j+1}(b, x, n, \mathcal{S})$ using the resource constraint:

$$b'_{j+1}(b, x, n, \mathcal{S}) = R\Big\{b + w(b, x, n, \mathcal{S}) + \hat{p}^s_j(b, x, n, \mathcal{S})\Big(1 - \mathbf{1}\,\phi\Big) + \mathbf{1}\,\phi\,x + l_j(b, x, n, \mathcal{S}) - f(l_j(b, x, n, \mathcal{S})) \\ - c_{j+1}(b, x, n, \mathcal{S}) - \hat{p}^s_j(b, x, n, \mathcal{S}) - \hat{p}^x_j(b, x, n, \mathcal{S})\,x'(b, x, n, \mathcal{S})\Big\} \tag{104}$$

- **Step 4**: Evaluate if the collateral constraint binds. If

$$\kappa_t\left(\hat{p}^s_j(b, x, n, \mathcal{S})\,(1 - \mathbf{1}_t\phi_t) + \mathbf{1}_t\phi_t\,x - l_j(b, x, n, \mathcal{S})\right) + \frac{b'_{j+1}(b, x, n, \mathcal{S})}{R_t} \geq 0 \tag{105}$$

the constraint does not bind and go to Step 6. Otherwise, the collateral constraint is binding and go to Step 5.

- **Step 5**: Solve for $b'_{j+1}(b, x, n, \mathcal{S})$ by evaluating the collateral constraint with equality:

$$b'_{j+1}(b, x, n, \mathcal{S}) = -R_t\Big\{\kappa_t\left(\hat{p}^s_j(b, x, n, \mathcal{S})\,(1 - \mathbf{1}_t\phi_t) + \mathbf{1}_t\phi_t\,x - l_j(b, x, n, \mathcal{S})\right)\Big\} \tag{106}$$

Determine $c_{j+1}(b, x, n, \mathcal{S})$ directly from the resource constraint:

$$c_{j+1}(b, x, n, \mathcal{S}) = b + w(b, x, n, \mathcal{S}) + \hat{p}^s_j(b, x, n, \mathcal{S})\Big(1 - \mathbf{1}\,\phi\Big) + \mathbf{1}\,\phi\,x \\ - \frac{b'_{j+1}(b, x, n, \mathcal{S})}{R} - \hat{p}^s_j(b, x, n, \mathcal{S}) - \hat{p}^x_j(b, x, n, \mathcal{S})\,x'(b, x, n, \mathcal{S}) + l_j(b, x, n, \mathcal{S}) - f(l_j(b, x, n, \mathcal{S})) \tag{107}$$

**Step 6**: Solve for $\mu_{j+1}(b, x, n, \mathcal{S})$ using the bonds Euler equation:

$$\mu_{j+1}(b, x, n, \mathcal{S}) = \Big(c_{j+1}(b, x, n, \mathcal{S}) + \nu_s g(1)\Big)^{-\psi} \\ - \beta R\,\mathbb{E}\left[\Big(c_j\Big(\hat{b}'_j(b, x, n, \mathcal{S}),\, x'(b, x, n, \mathcal{S}),\, \mathcal{S}'\Big) + \nu_s g(1)\Big)^{-\psi}\right] \tag{108}$$

- **Step 7**: Update the insurance and house price:

$$p^x_{j+1}(b, x, n, \mathcal{S}) = \frac{\beta\,\mathbb{E}_t\left[\mathbf{1}_{t+1}\,\phi_{t+1}\Big(\mathcal{U}_{j+1}\Big(\hat{b}'_{j+1}, x', n', \mathcal{S}'\Big)^{-\psi} + \mu_{j+1}(\hat{b}'_{j+1}, x', \mathcal{S}')\,\kappa'\Big)\right]}{\mathcal{U}_{j+1}(b, x, n, \mathcal{S})^{-\psi}} \tag{109}$$

$$
\begin{aligned}
&p^s_{j+1}(b,x,n,\mathcal{S}) \\
&= \beta\,\mathbb{E}_t\left[\frac{\mathcal{U}_{j+1}\left(\hat{b}'_{j+1},x',n',\mathcal{S}'\right)^{-\psi}}{\mathcal{U}_{j+1}(b,x,n,\mathcal{S})^{-\psi}}\left\{\nu_s g'(1)+\hat{p}^s_j(\hat{b}'_{j+1}(b,x,n,\mathcal{S}),\,x'(b,x,n,\mathcal{S}),\,\mathcal{S}')(1-\mathbf{1}_{t+1}\phi_{t+1})\right\}\right. \\
&\left.+\frac{\mu_{j+1}(\hat{b}'_{j+1},x',\mathcal{S}')\,\kappa'}{\mathcal{U}_{j+1}(b,x,n,\mathcal{S})^{-\psi}}\left\{\hat{p}^s_j(\hat{b}'_{j+1}(b,x,n,\mathcal{S}),\,x'(b,x,n,\mathcal{S}),\,\mathcal{S}')(1-\mathbf{1}_{t+1}\phi_{t+1})\right\}\right]
\end{aligned}
\tag{110}
$$

where

$$
\mathcal{U}_{j+1}\left(\hat{b}'_{j+1},x',n',\mathcal{S}'\right)=c_{j+1}\left(\hat{b}'_{j+1}(b,x,n,\mathcal{S}),x'(b,x,n,\mathcal{S}),\mathcal{S}'\right)+\nu_s g(1) \tag{111}
$$

$$
\mathcal{U}_{j+1}(b,x,n,\mathcal{S})=c_{j+1}(b,x,n,\mathcal{S})+\nu_s g(1) \tag{112}
$$

$$
\tag{113}
$$

- **Step 8**: Compute the difference between the old and new insurance prices. Update the insurance price if the difference is not smaller than a lower than an upper bound, and return to step 0.

## 10.6 Computational Algorithm - planner's equilibrium

The algorithm to solve the planner's problem is similar to Bianchi and Mendoza (2018), but extends with insurance intermediaries.

- **Step 0**: Start iteration $j$ with an initial conjecture of the pricing functions $\hat{p}^x_j(b,x,n,\mathcal{S})$, $\hat{p}^s_j(b,x,n,\mathcal{S})$, decision rule for bonds $\hat{b}'_j(b,x,n,\mathcal{S})$, insurance $\hat{x}'_j(b,x,n,\mathcal{S})$, consumption $\hat{c}_j(b,x,n,\mathcal{S})$, wages $\hat{w}_j(b,x,n,\mathcal{S})$, the multiplier $\hat{\mu}_j(b,x,n,\mathcal{S})=0$ and initial value function $\hat{V}_j(b,x,n,\mathcal{S})$. I use as initial functions the policies of the decentralized equilibrium.

- **Step 1**: Using the guess $\hat{p}^x_j$, compute insurance supply, wages and return on assets by iterating on the insurer problem.

- **Step 2**: Given $\hat{V}_j$, $\hat{x}'_j$, $\hat{p}^s_j$, $\hat{w}_j$, solve for the value and policy functions:

$$
\mathcal{V}(b,x,n,\mathcal{S})=\max_{c,b',s,x',\mu}u(c,s)+\beta\,\mathbb{E}_{\mathcal{S}'|\mathcal{S}}\mathcal{V}(b',x',n',\mathcal{S}') \tag{114}
$$

subject to

$$
c+\frac{b'}{R}+p^x x' p^r=b+hw+\mathbf{1}\phi\left[xp^r-p^s\right]+l-f(l) \tag{115}
$$

$$n' = a(1 + R'^k) - \mathbf{1}'\phi' x' p^r \tag{116}$$

$$\psi n = x' \tag{117}$$

$$\frac{b'}{R} \geq -\kappa \left[ p^s \left(1 - \mathbf{1}\phi\right) + \mathbf{1}\phi x - l \right] \tag{118}$$

$$\mu \left( \frac{b'}{R} + \kappa \left[ p^s \left(1 - \mathbf{1}\phi\right) + \mathbf{1}\phi x - l \right] \right) = 0 \tag{119}$$

$$\begin{aligned} p^s \, u'(c,1) = \beta \, \mathbb{E} \Bigg[ & u'\Big(c'(b', x', n', \mathcal{S}'), 1\Big) \\ & \times \left\{ \nu_s g'(1) + p^{s'}(b', x', n', \mathcal{S}') \left(1 - \mathbf{1}'\phi'\right) \right\} \\ & + p^{s'}(b', x', n', \mathcal{S}') \left(1 - \mathbf{1}'\phi'\right) \, \mu'(b', x', n', \mathcal{S}') \, \kappa' \Bigg] \end{aligned} \tag{120}$$

This problem is solved using value function iteration with linear interpolation when the insurance, bond and insurer net worth positions are outside the grid. First assume the collateral constraint is not binding. Use a Newton type algorithm to find the candidate solutions for $\hat{b}'_{j+1}$ and $\hat{x}'_{j+1}$. Solve for $\hat{c}_{j+1}$ using the budget constraint. I solve for policies using a smooth optimizer. The key distinction with the decentralized equilibrium is that the planner endogenously updated insurer net worth in the next-period's value function.

- **Step 3**: Check if the collateral constraint (118) is binding. If binding, solve for each possible $b'_{j+1}$, the combinations $\mu$, $c$, $p^s$ and $p^x$ that make hold the conditions hold with equality. Pick the combination that has the maximum value.

- **Step 4**: Repeat steps 1-3 until all functions convergence.